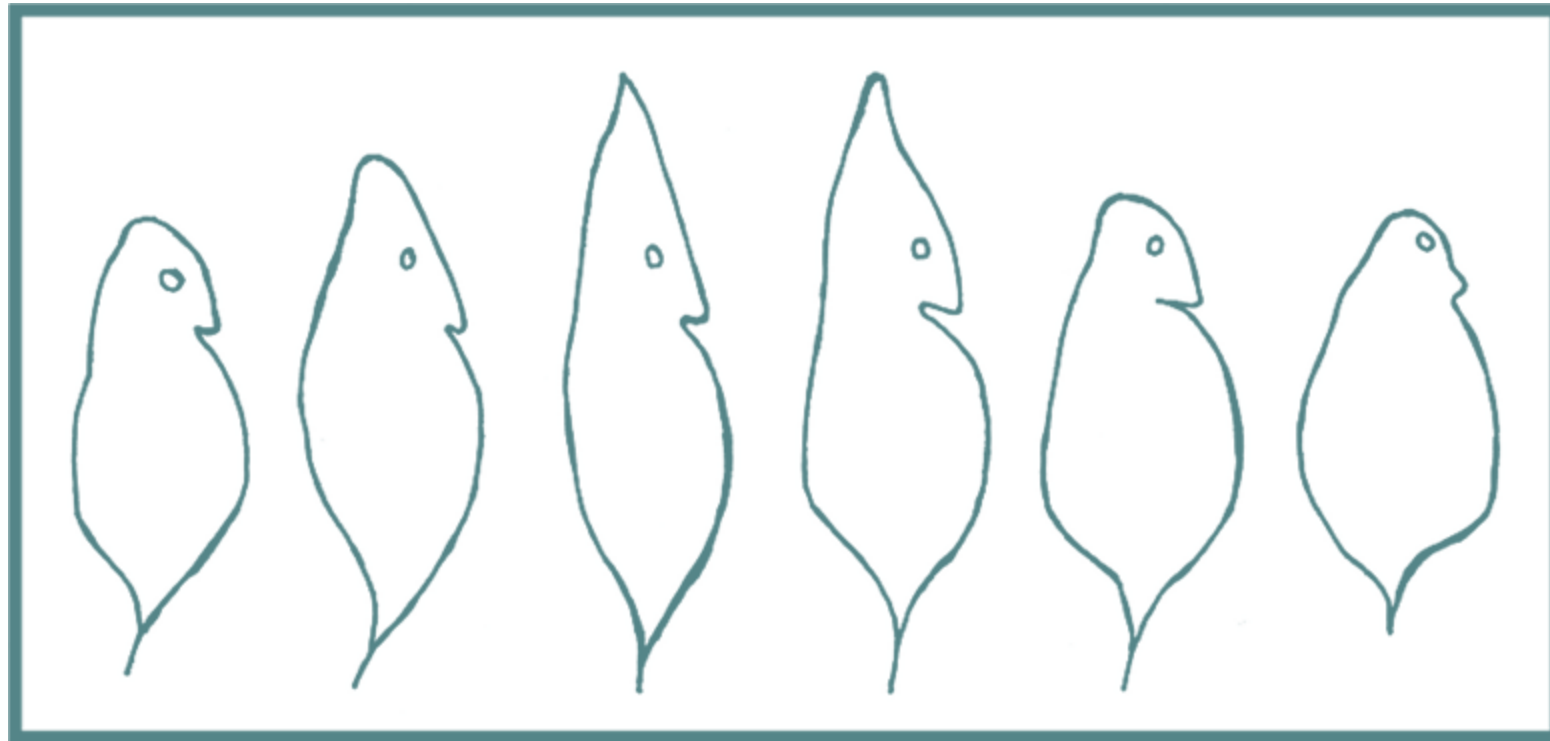


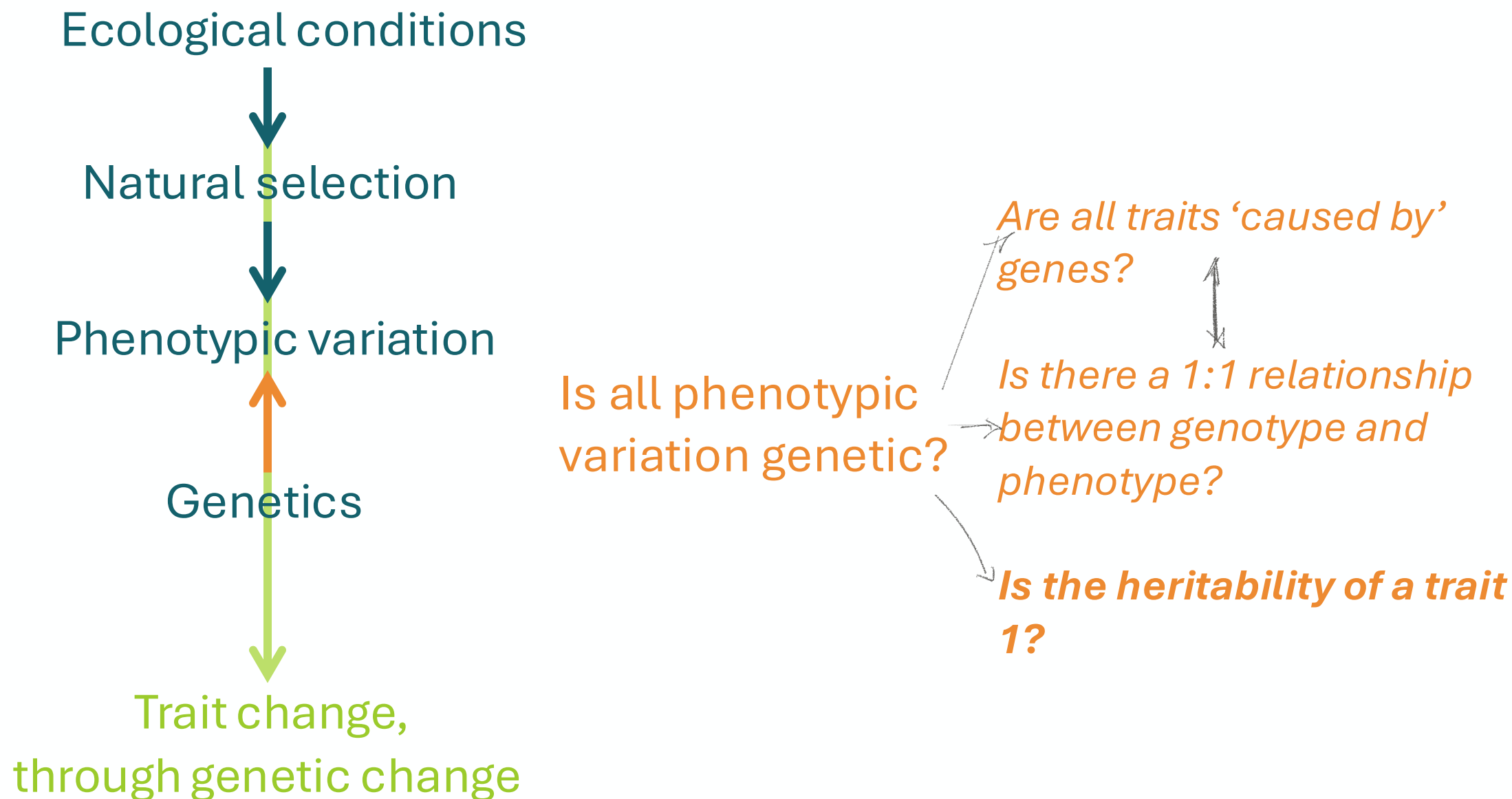


Genotype $\neq$ phenotype

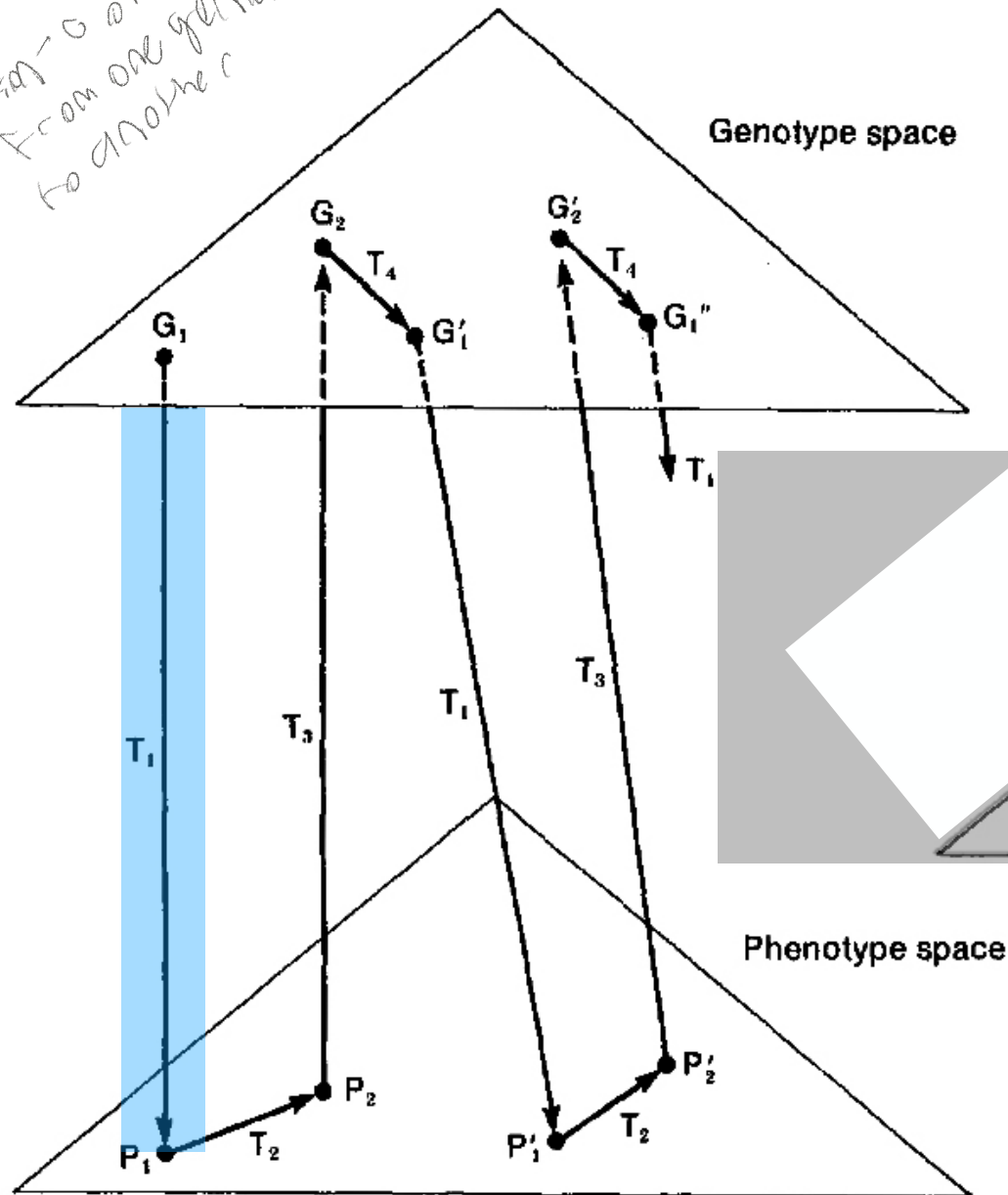
Dr. Laura van Holstein

lavanho@emory.edu

Room 215



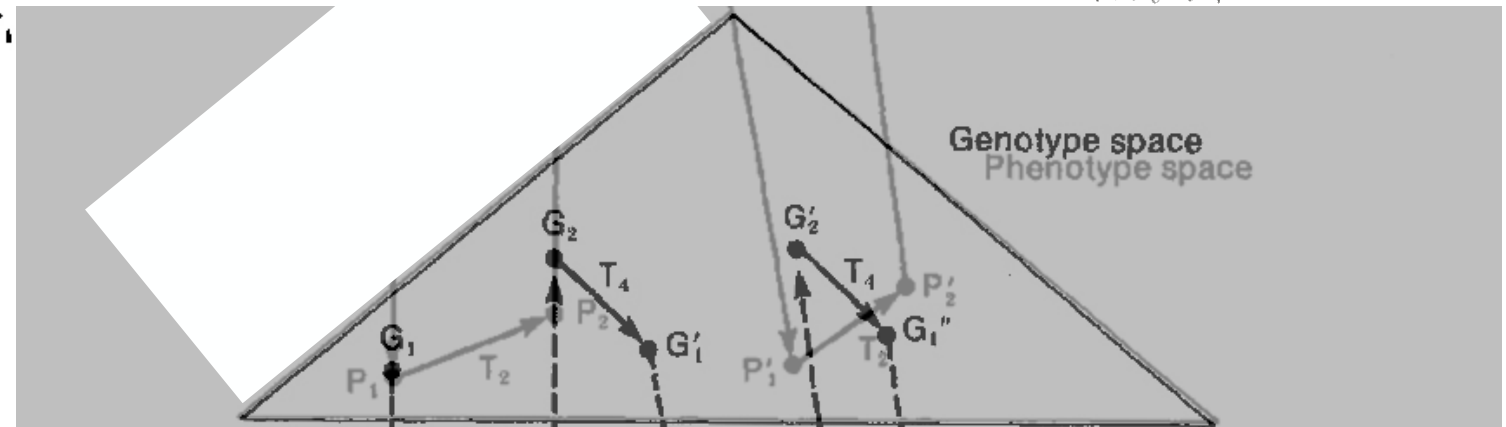
Evolution - going from one genotype to another



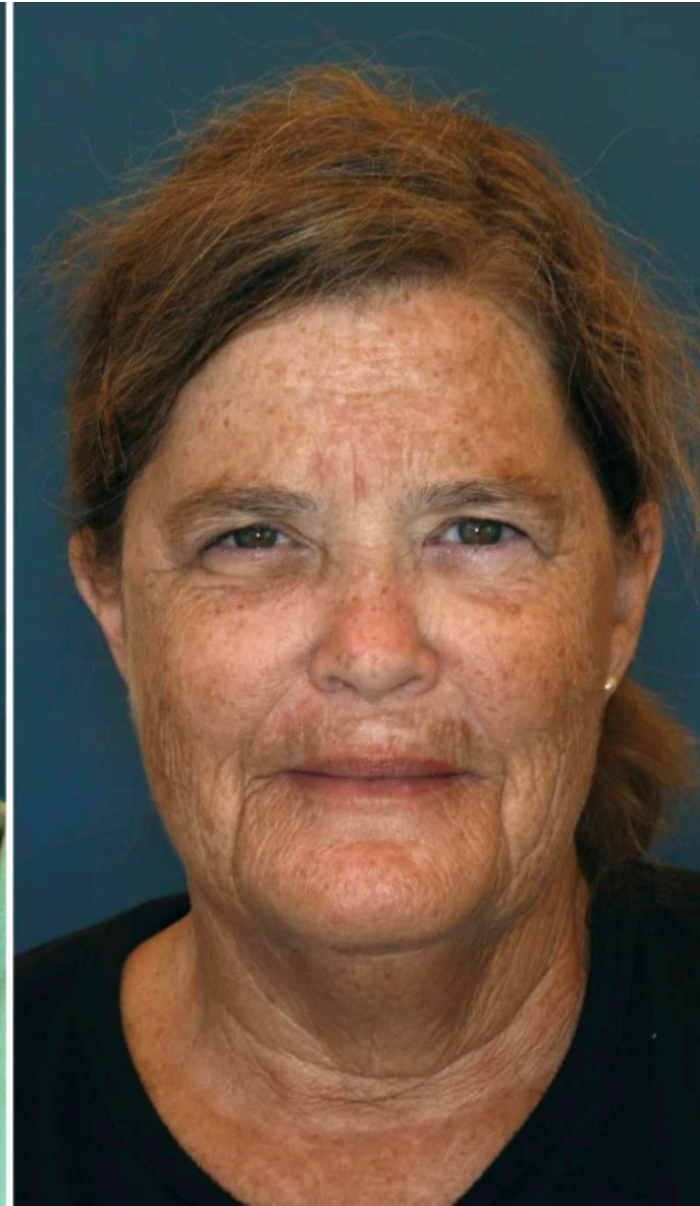
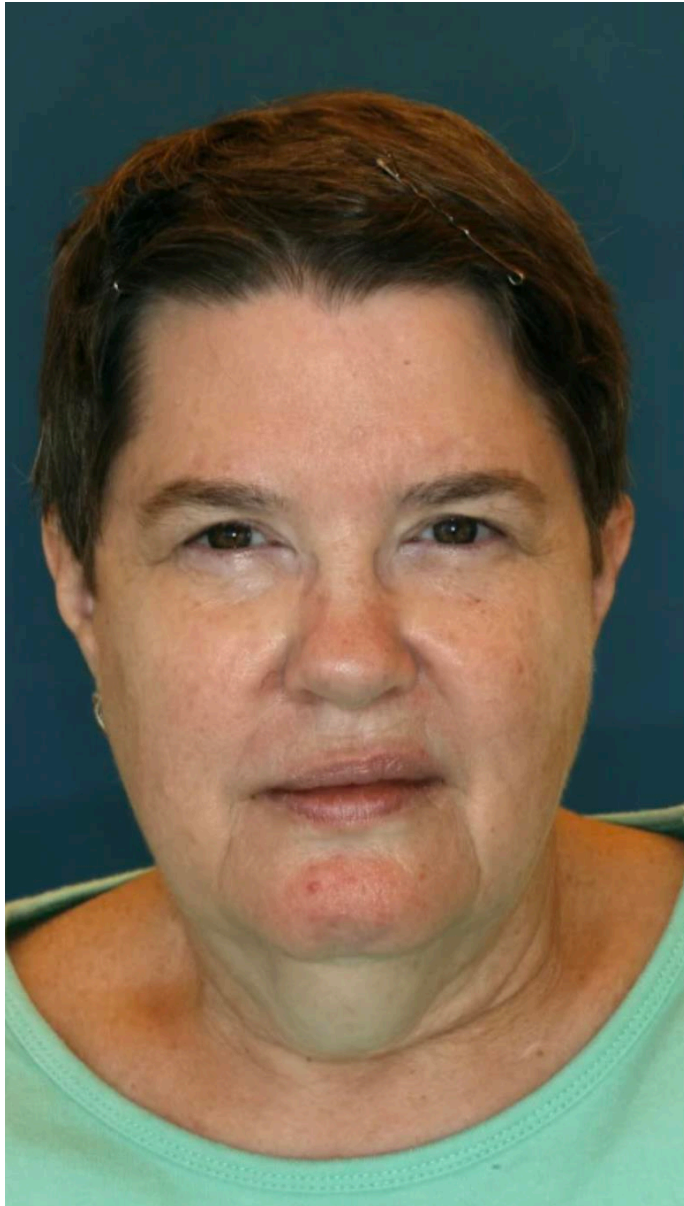
Heritability is not always 1
Est for classical markers is what's 0.25?

If genotype mapped perfectly on to phenotype, $G_1 = P_1$:

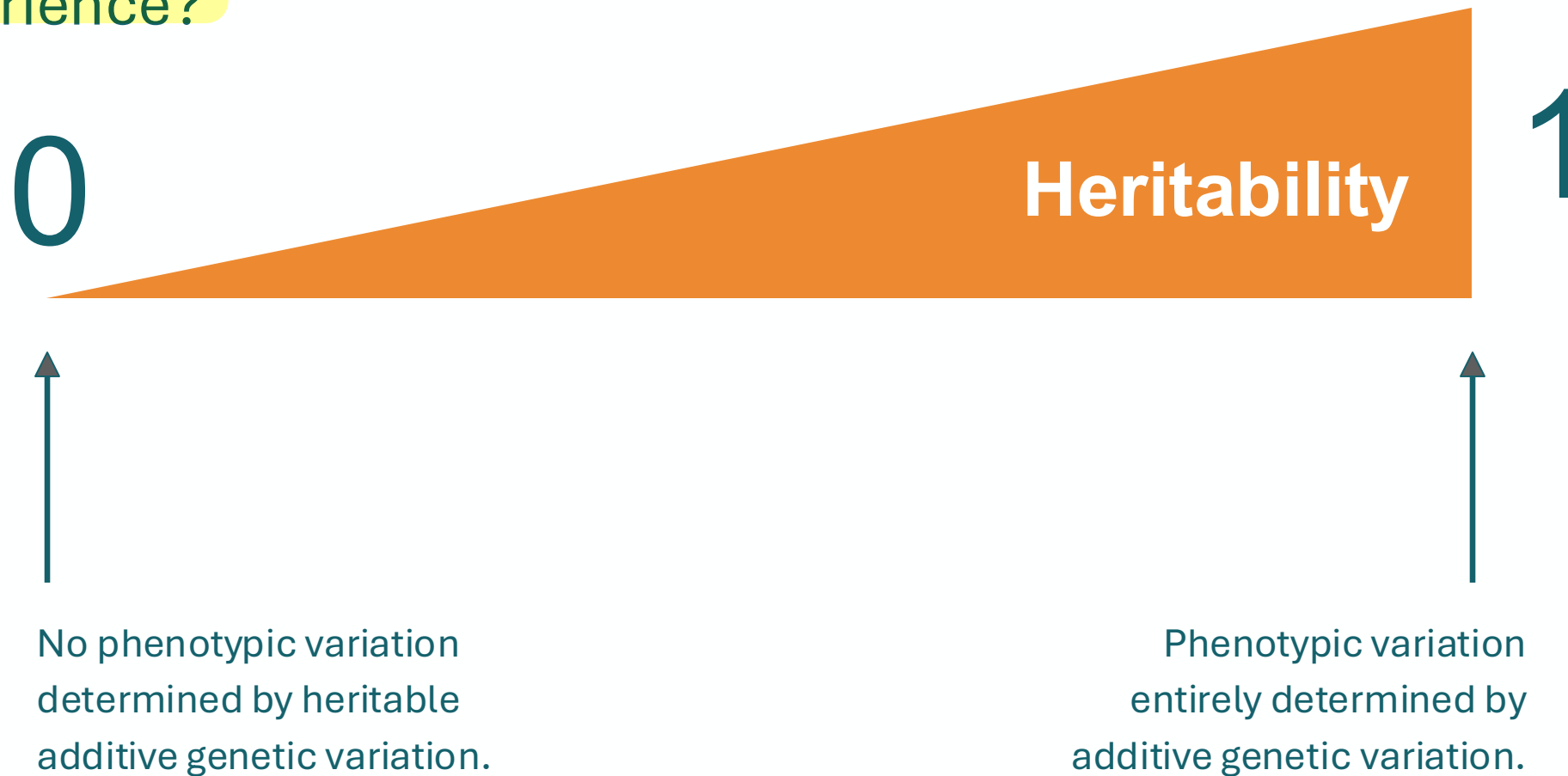
↳ They would be exactly the same



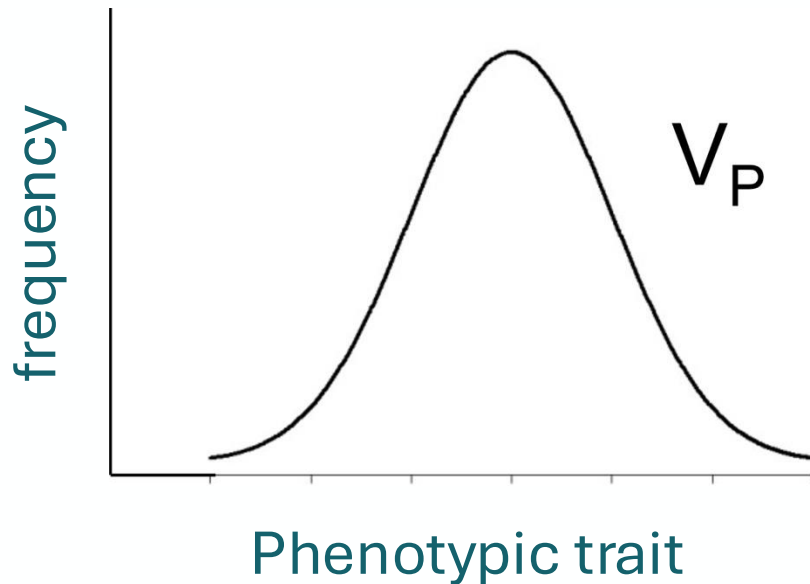
Monozygotic
twins
- have
the
same
genotype
(have the same
G) but
have different
phenotypes



Heritability:
What **proportion** of the phenotypic variation between **individuals in a population** is due to their **genetic variation** versus **variation in the environment** they experience?



$$V_P = V_G + V_E + V_{G \times E}$$



V_P = phenotypic variance

= a statistical measure of the variation in phenotype across all the **individuals in a population**

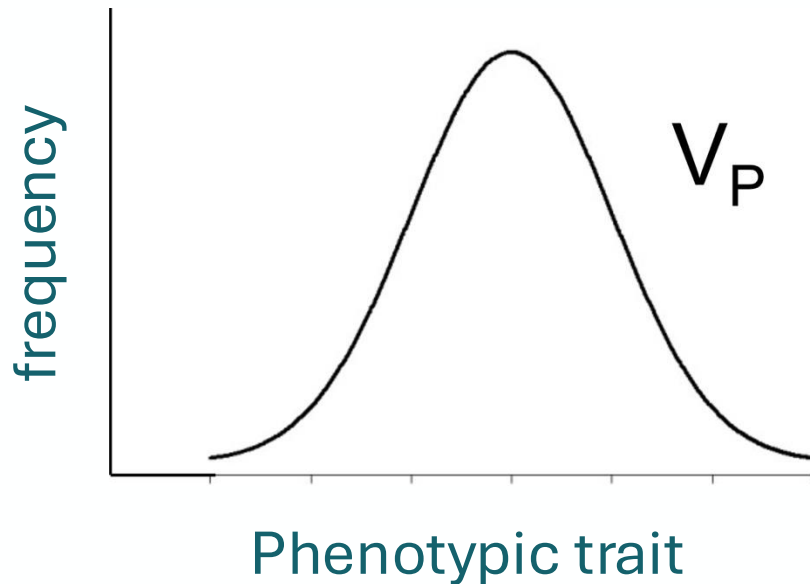
V_G = genotypic variance

= the portion of V_P that is due to variation in the **genotype** between individuals in the population

V_E = environmental variance

= the portion of V_P that is due to variation in the **environment** experienced by individuals in the population

$$V_P = V_G + V_E + V_{G \times E}$$



Subs of V_G ?
 V_A = additive genetic variance

= the portion of V_P that is due to the independent effects of single alleles on the phenotype and that is therefore **passed onto the next generation**

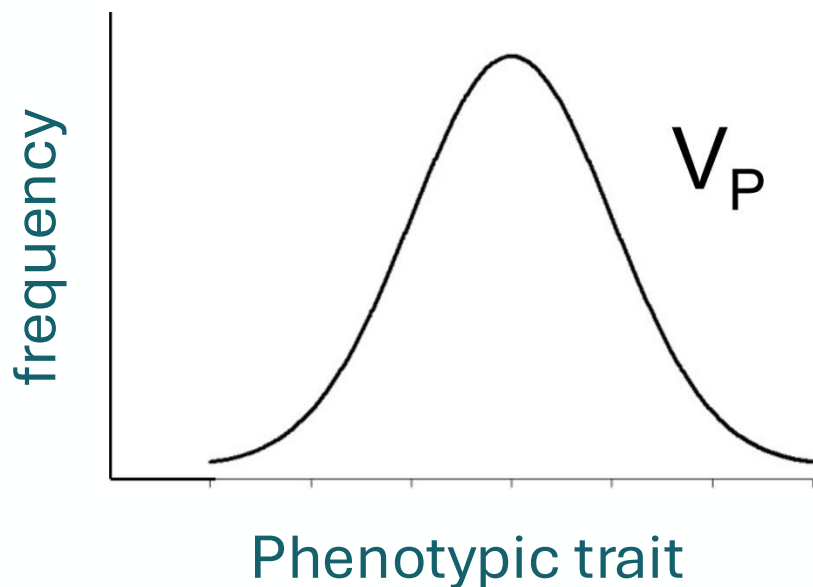
V_D = dominance variance

= the portion of V_P that is due to the dominant/recessive interactions between alleles at a single locus [these interactions are broken up by meiosis so are not passed on to the next generation]

V_I = interaction variance

= the portion of V_P that is due to the epistatic interactions between alleles at different loci in the genome [these interactions are broken up by meiosis so are not passed on to the next generation]

$$V_P = V_G + V_E + V_{G \times E}$$



V_A = additive genetic variance

= the portion of V_P that is due to the independent effects of single alleles on the phenotype and that is therefore **passed onto the next generation**

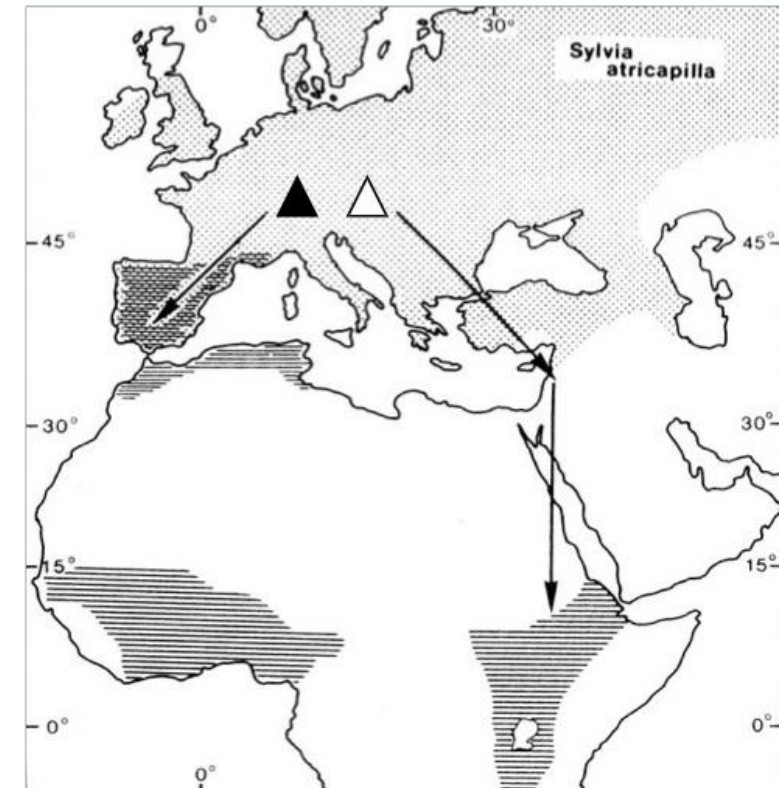
Narrow-sense heritability

$$h^2 = \frac{V_A}{V_P} = \frac{V_A}{V_G + V_E}$$

- varies from 0 to 1
- quantifies the resemblance between offspring and parents
- is NOT an intrinsic property of a trait — it is an estimate specific to the population in which it was measured, at that time, in that place
- predicts the response of a trait to selection on that trait

Blackcap migration comprises several highly heritable behavioural traits

Migrate south in the winter but there are 2 routes
The birds split into 2 populations
migration time varies

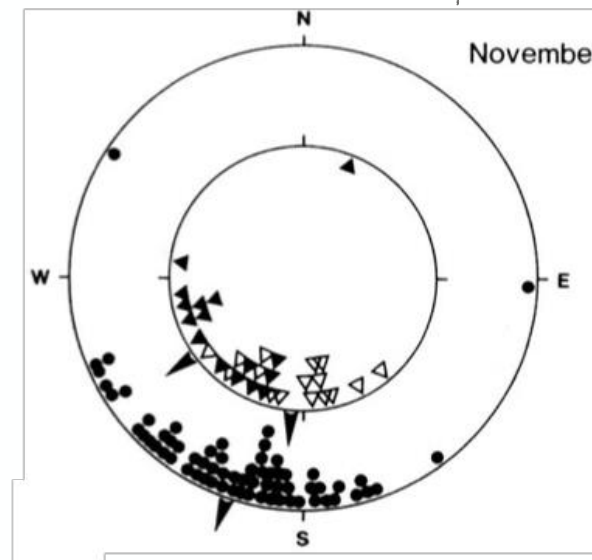
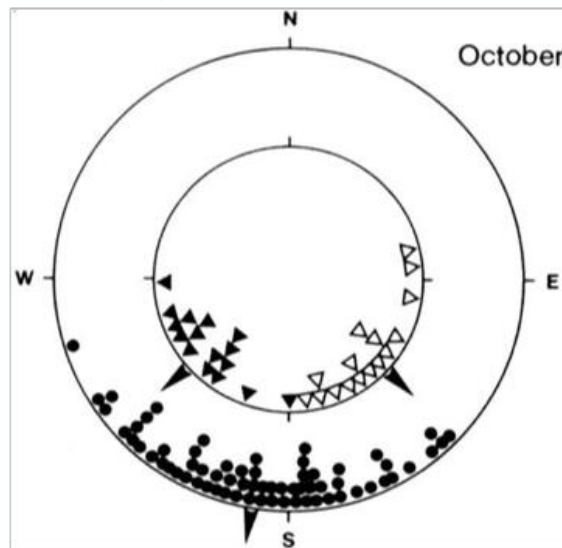


Two overwintering routes from central Europe, differing in:

- Date of migration onset
- Duration of migration
- Direction of migration

Migratory traits can be quantified in the lab by placing birds in chambers

↳ No info about day length or anything for what time it is - but birds in chambers would try to fly in a certain direction



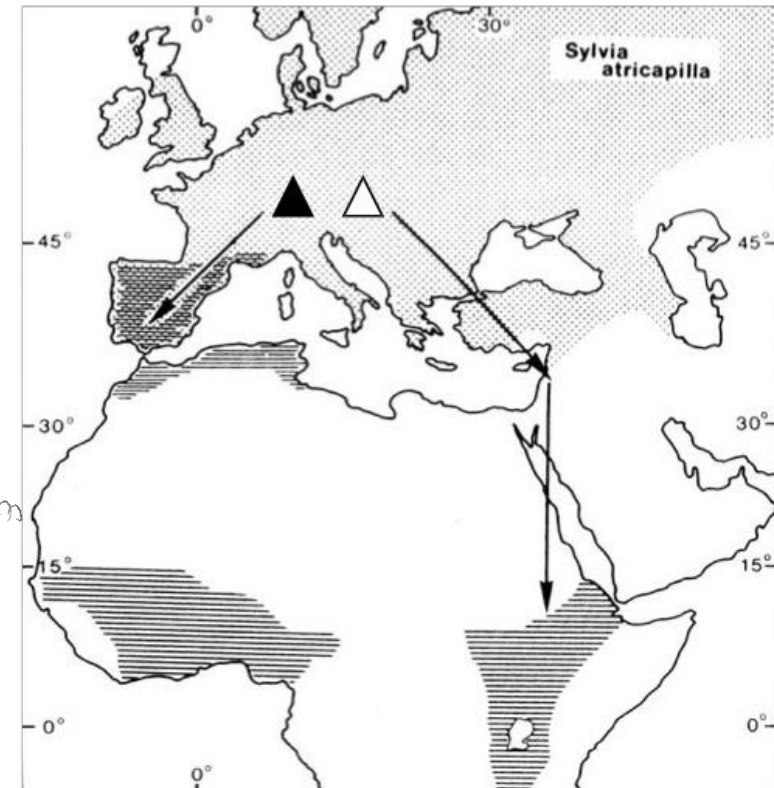
▲ SW migrating

△ SE migrating

● Hybrids



Berthold (1996)

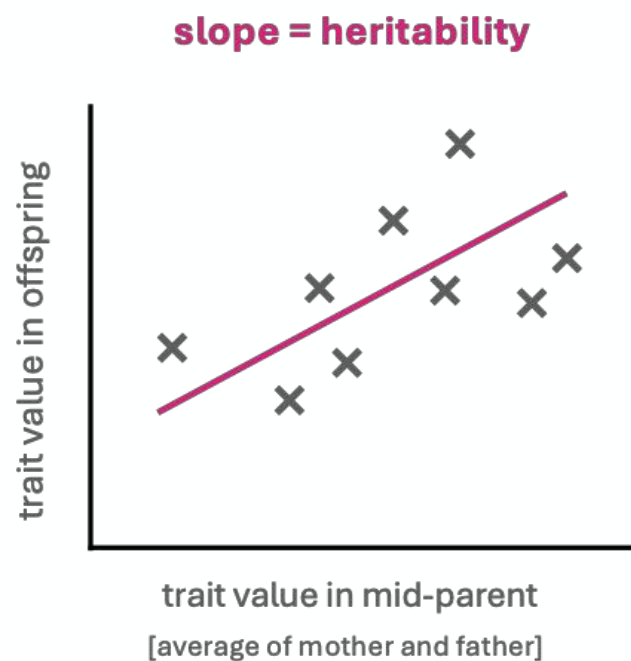


Two overwintering routes from central Europe, differing in:

- Date of migration onset
- Duration of migration
- Direction of migration

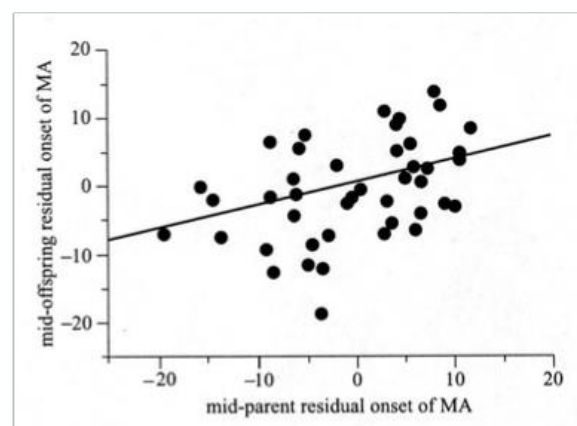
Estimating heritability of blackcap migratory traits

Method 1: Mid-parent offspring regression



Ideally, you would cross-foster offspring to control for any correlation between parental genotype and rearing environment.

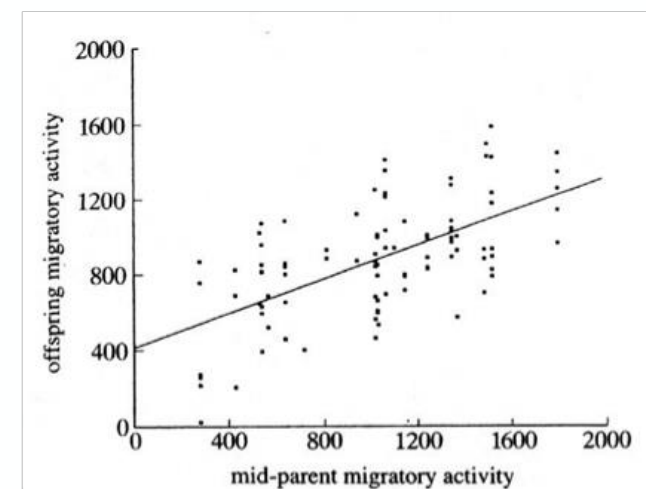
Trait:
Date of onset of migration



$$h^2 = 0.34 - 0.45$$

around 40% of phenotype
WAT this trait is heritable?

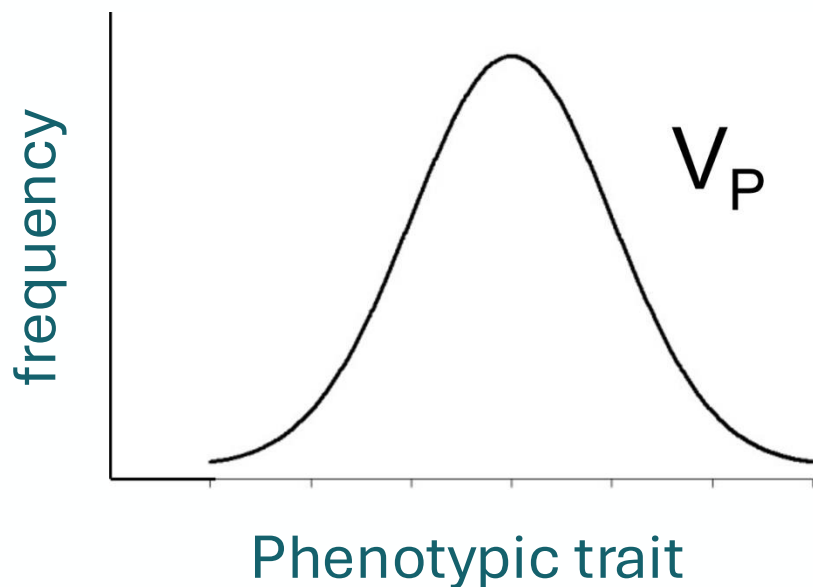
Trait:
Duration of migratory activity



$$h^2 = 0.37 - 0.46$$

Berthold (1996)

$$V_P = V_G + V_E + V_{G \times E}$$



V_A = additive genetic variance

= the portion of V_P that is due to the independent effects of single alleles on the phenotype and that is therefore **passed onto the next generation**

Variation may be underpinned by variation at many genetic loci ('**polygenic**') or at a **few** (or single ones!)



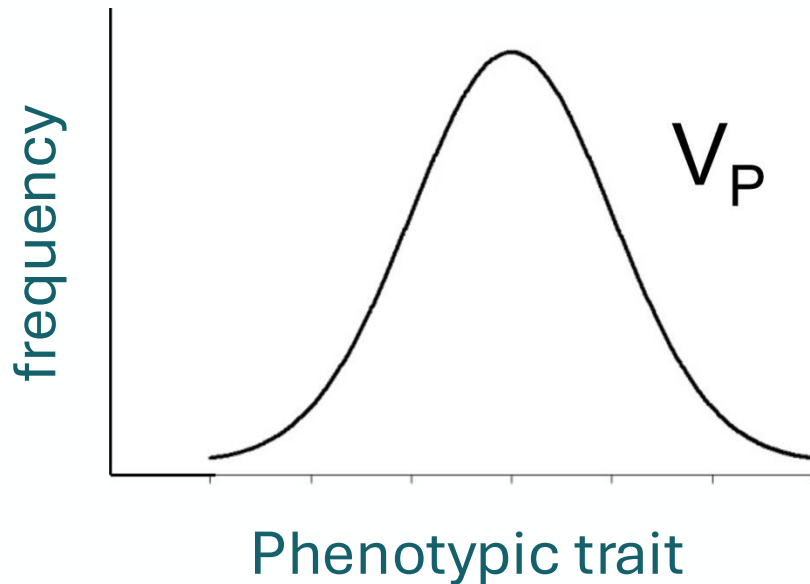
Single gene: Large effect each



Polygenic: Small effect each

most phenotypic traits are hard to define
are complex (polygenic)

$$V_P = V_G + V_E + V_{G \times E}$$



V_A = additive genetic variance

= the portion of V_P that is due to the independent effects of single alleles on the phenotype and that is therefore **passed onto the next generation**

Variation may be underpinned by variation at many genetic loci (**polygenic**) or at a few (or single ones!)



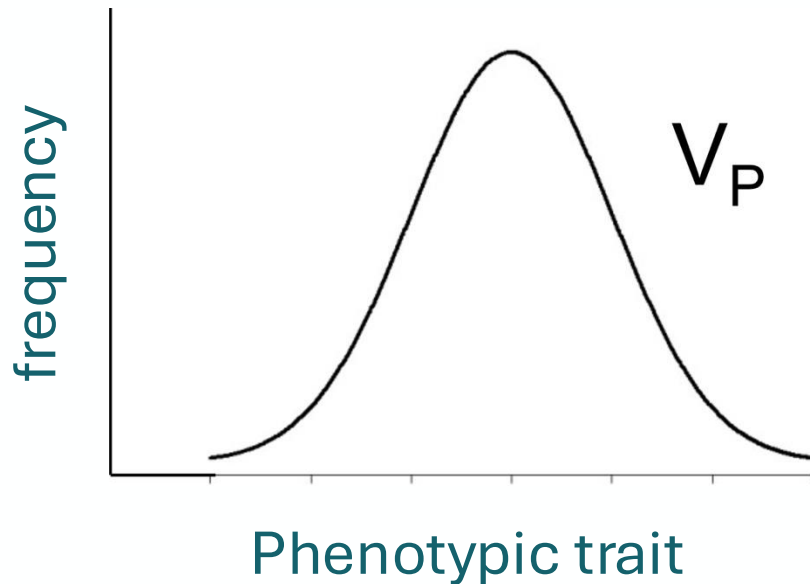
Single gene: Large effect each



Polygenic: Small effect each

Most phenotypic 'traits' are polygenic, or **complex**.

$$V_P = V_G + V_E + V_{G \times E}$$



V_P = phenotypic variance

= a statistical measure of the variation in phenotype across all the **individuals in a population**

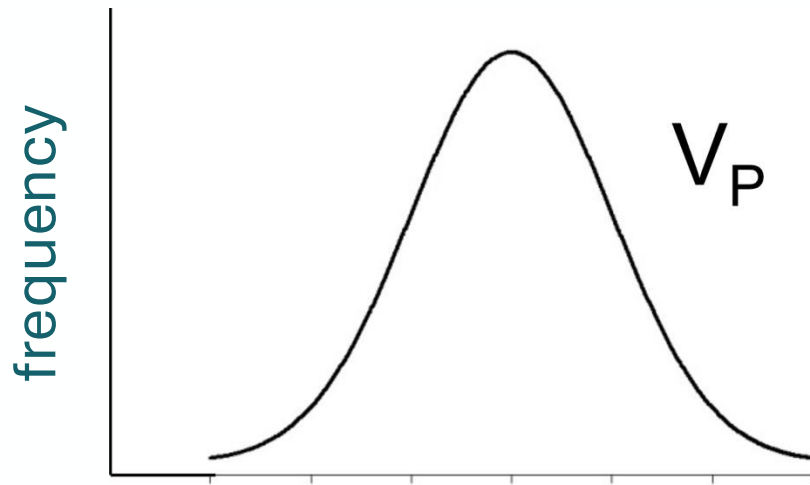
V_G = genotypic variance

= the portion of V_P that is due to variation in the genotype between individuals in the population

V_E = environmental variance

= the portion of V_P that is due to variation in the environment experienced by individuals in the population

$$V_P = V_G + V_E + V_{G \times E}$$



Phenotypic trait

It is hard to think of phenotypic traits solely caused by environment

$V_{G \times E}$ = genotype-environment interactions

= the portion of V_P that is due to **variation in the response of genotypes to the environment**

V_P = phenotypic variance

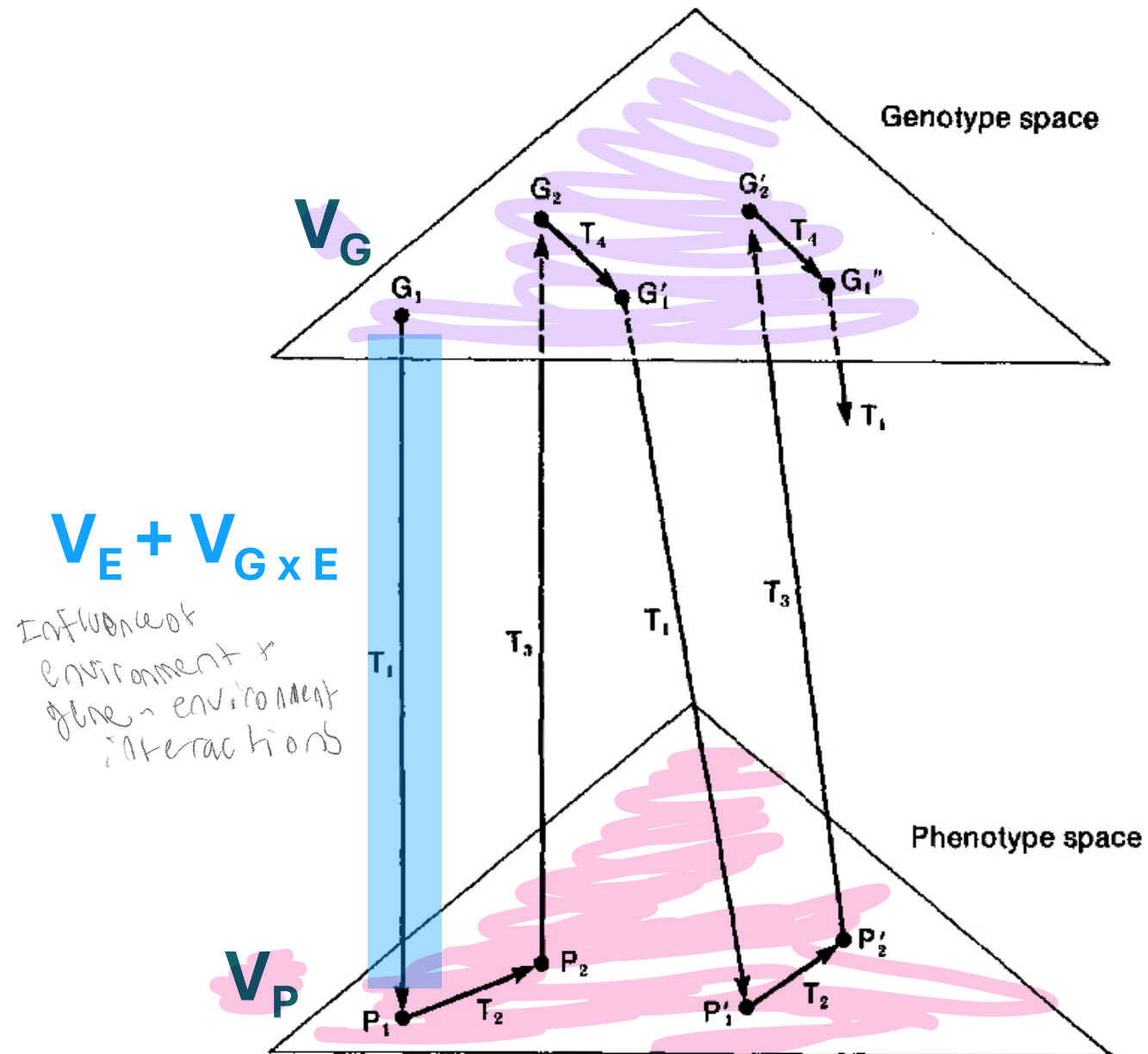
= a statistical measure of the variation in phenotype across all the **individuals in a population**

V_G = genotypic variance

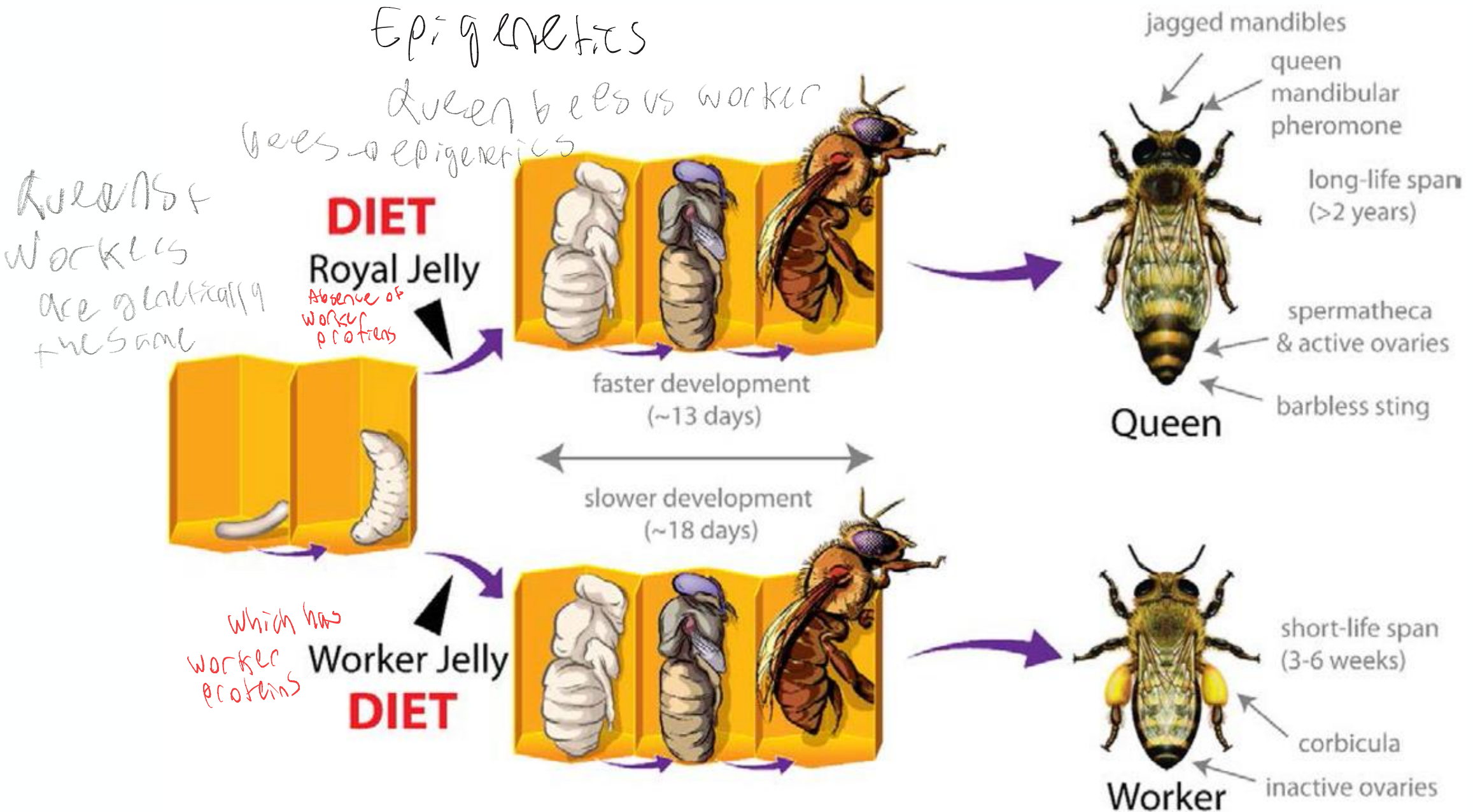
= the portion of V_P that is due to variation in the genotype between individuals in the population

V_E = environmental variance

= the portion of V_P that is due to variation in the environment experienced by individuals in the population

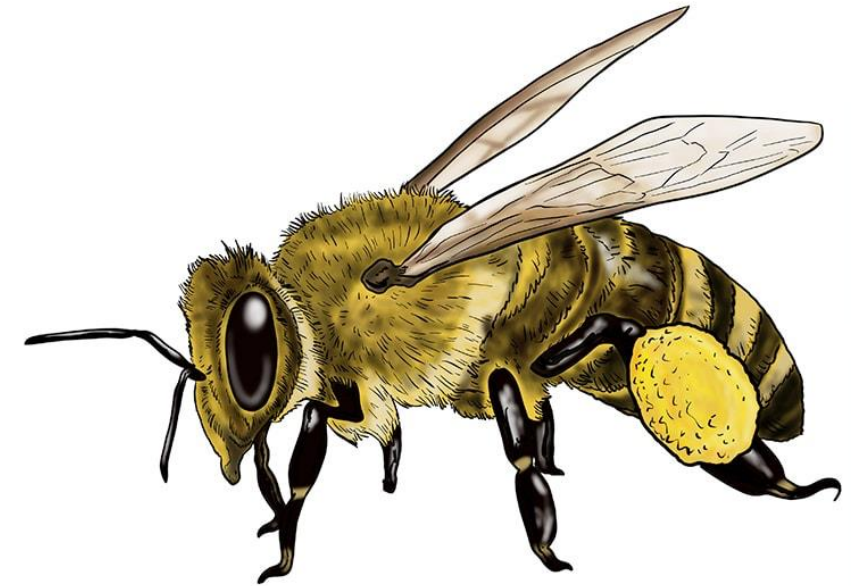


- Types of Environment + Gene-environment interactions
- 1 Epigenetics
 - 2 Reaction norms



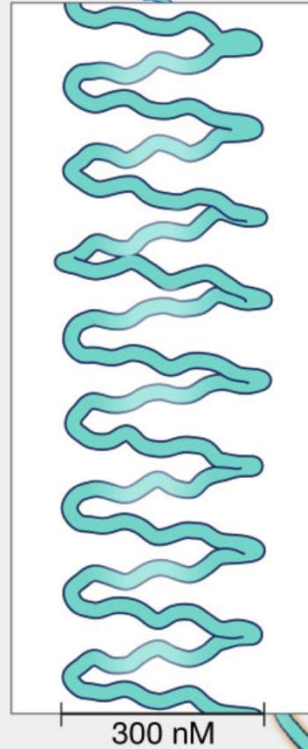
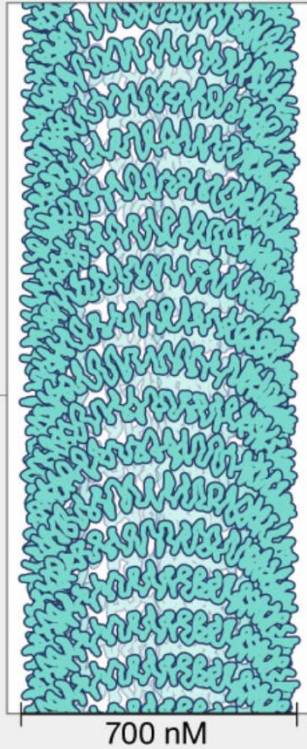
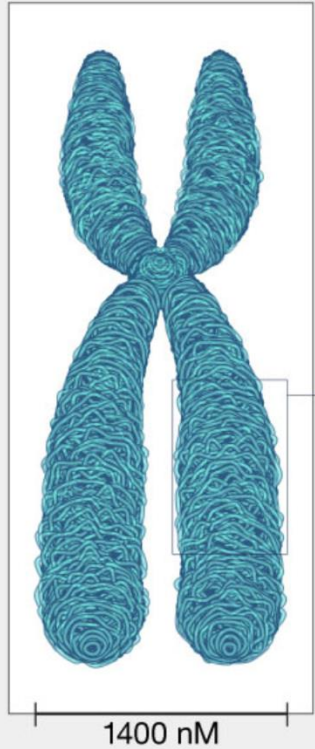
Epigenetics

- Superimposed upon the DNA sequence is a layer of **heritable “epigenetic” information**
 - **Chemical modifications** to cytosine bases and to the histone proteins that package the genome.
- By **regulating chromatin structure and DNA accessibility**, these chemical changes influence how the genome is expressed as phenotype.

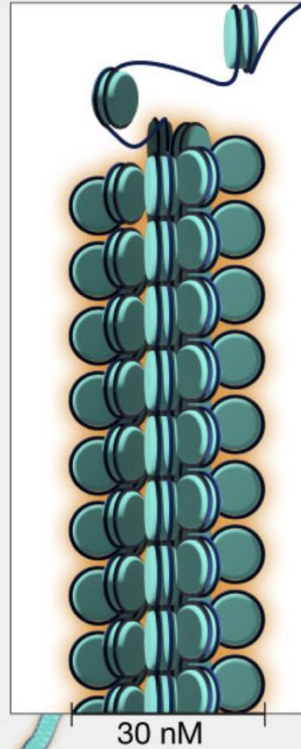


Regulate what
can be transcribed or translated,
either upregulating or
downregulating gene expression

Metaphase chromosome

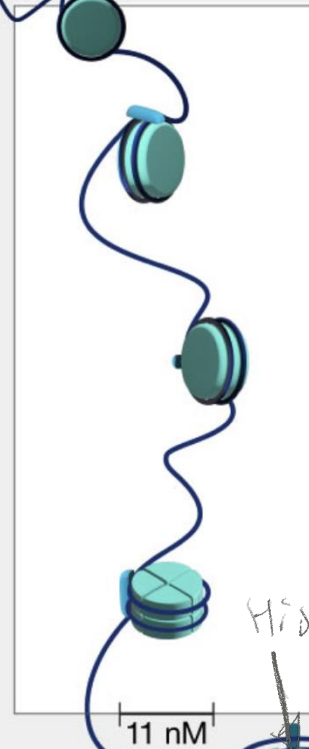


Chromatin fiber

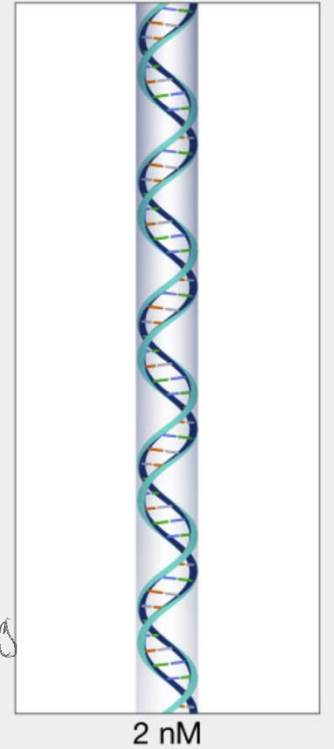


DNA coil around histones to make nucleosomes

Nucleosomes



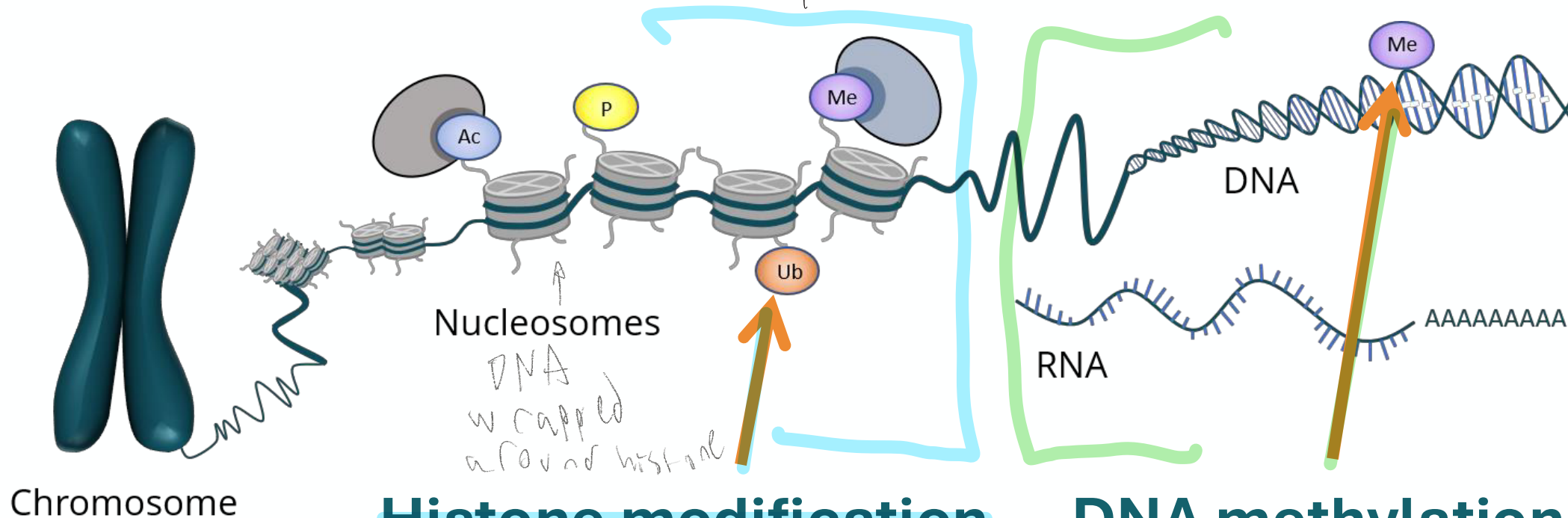
DNA



Nucleosomes coiled to make chromatin

Histone

Epigenetics w/AT this process



Histone modification

- >100 different posttranslational modifications (acetylation, methylation, phosphorylation, and ubiquitination...)
- *Histone tails* – attachment point that affects wrapping

Can be associated with increased or decreased transcription

DNA methylation

- Methylation of cytosine in CpG dinucleotides

Generally associated with decreased transcription

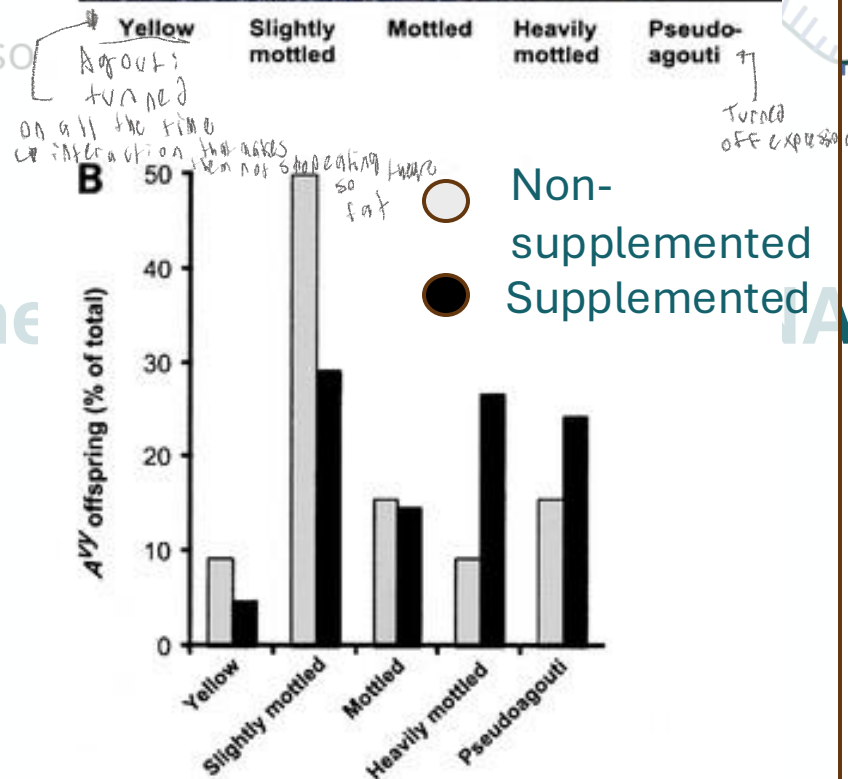
Mice -

Maternal diet that promotes **methylation** of the A^y gene leads to dark coats and healthier babies

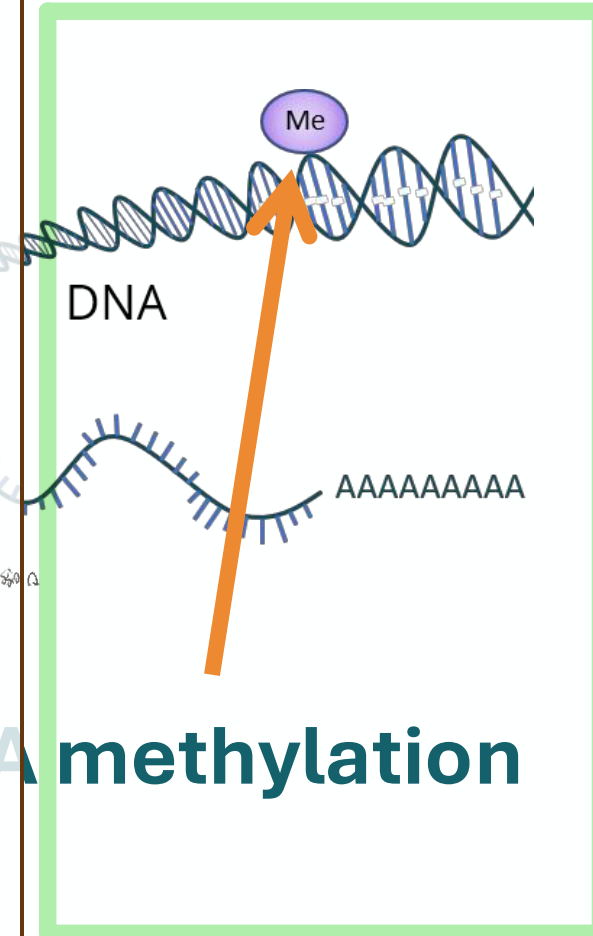
Agouti gene can be turned on or off

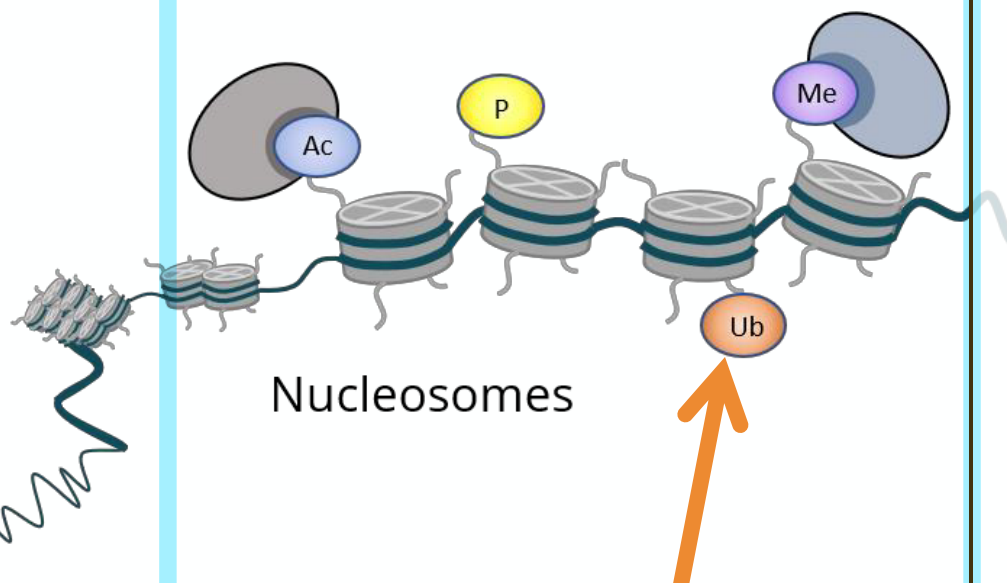


increased diet that promotes methylation of agouti gene increases agouti expression?



Waterland & Jirtle, 2003





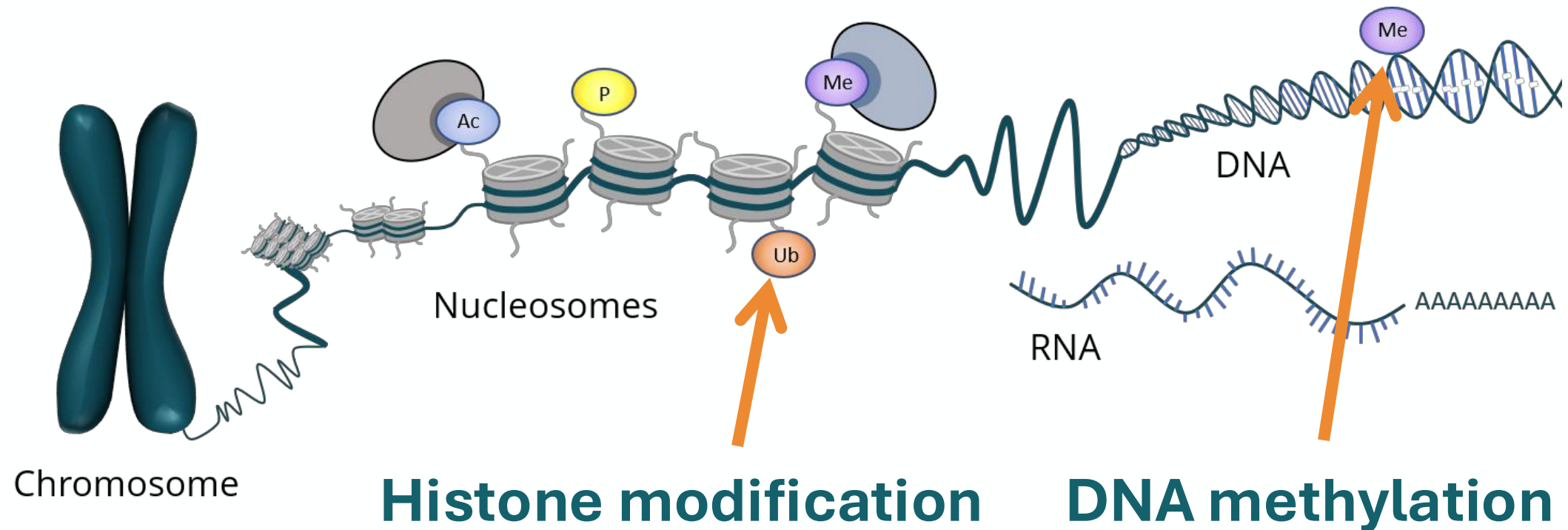
Histone modification

Red-eared sliders



- **KDM6B** (protein) promotes transcription of **male sex-determining gene Dmrt1**
- **Eliminates** methylation of a histone close to promotor of gene **Dmrt1**
- In male-producing temperatures (26C), eliminating KDM6B triggers male-to-female sex reversal

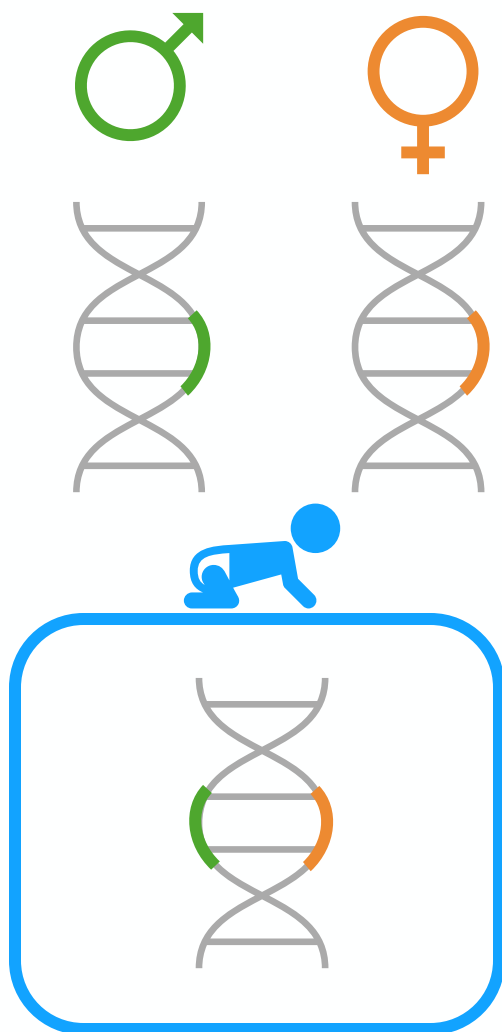
Ge et al. (2018)



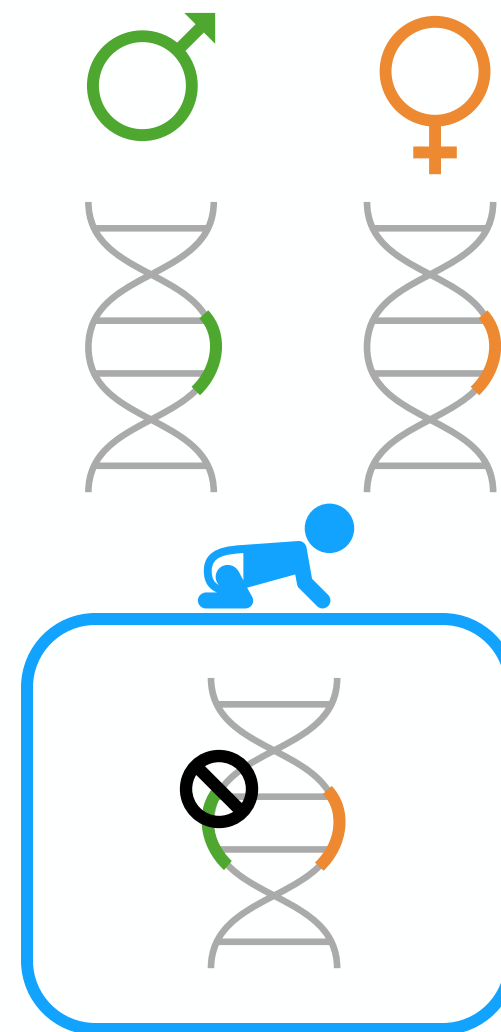
Both can underlie genomic imprinting.

Genomic imprinting: parent-specific expression of a gene

Mendelian

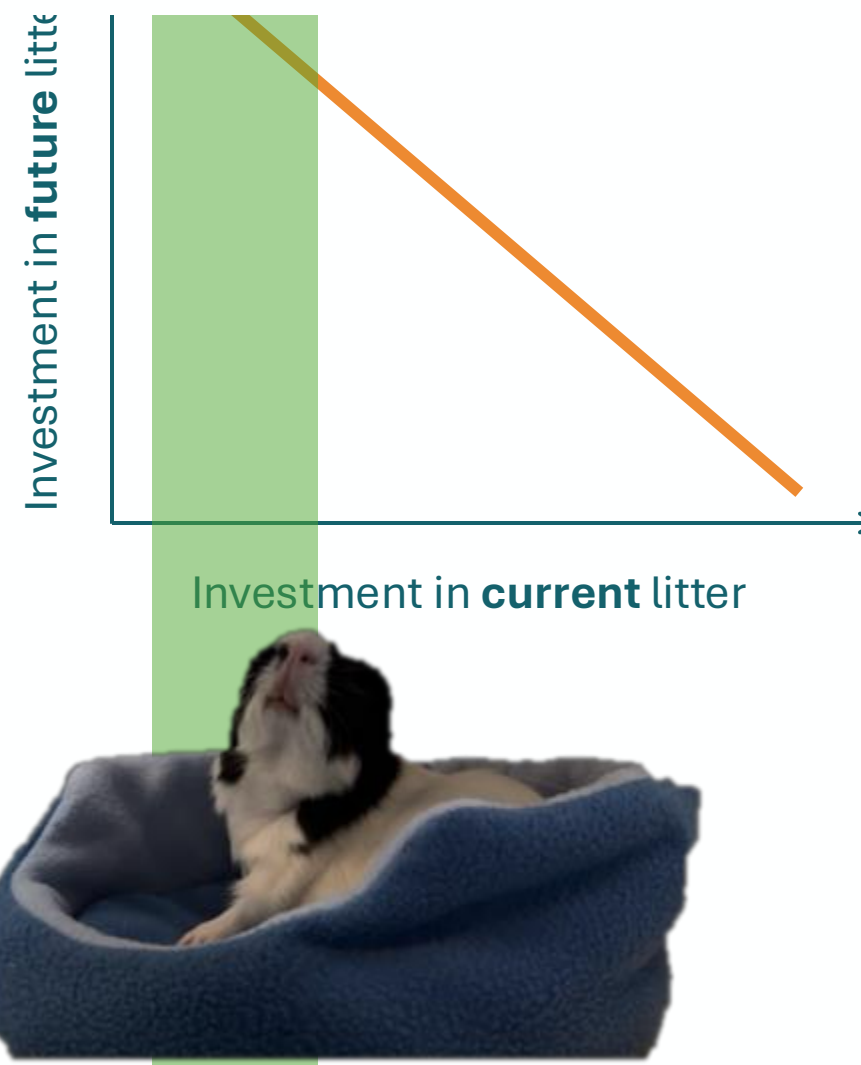


Imprinting

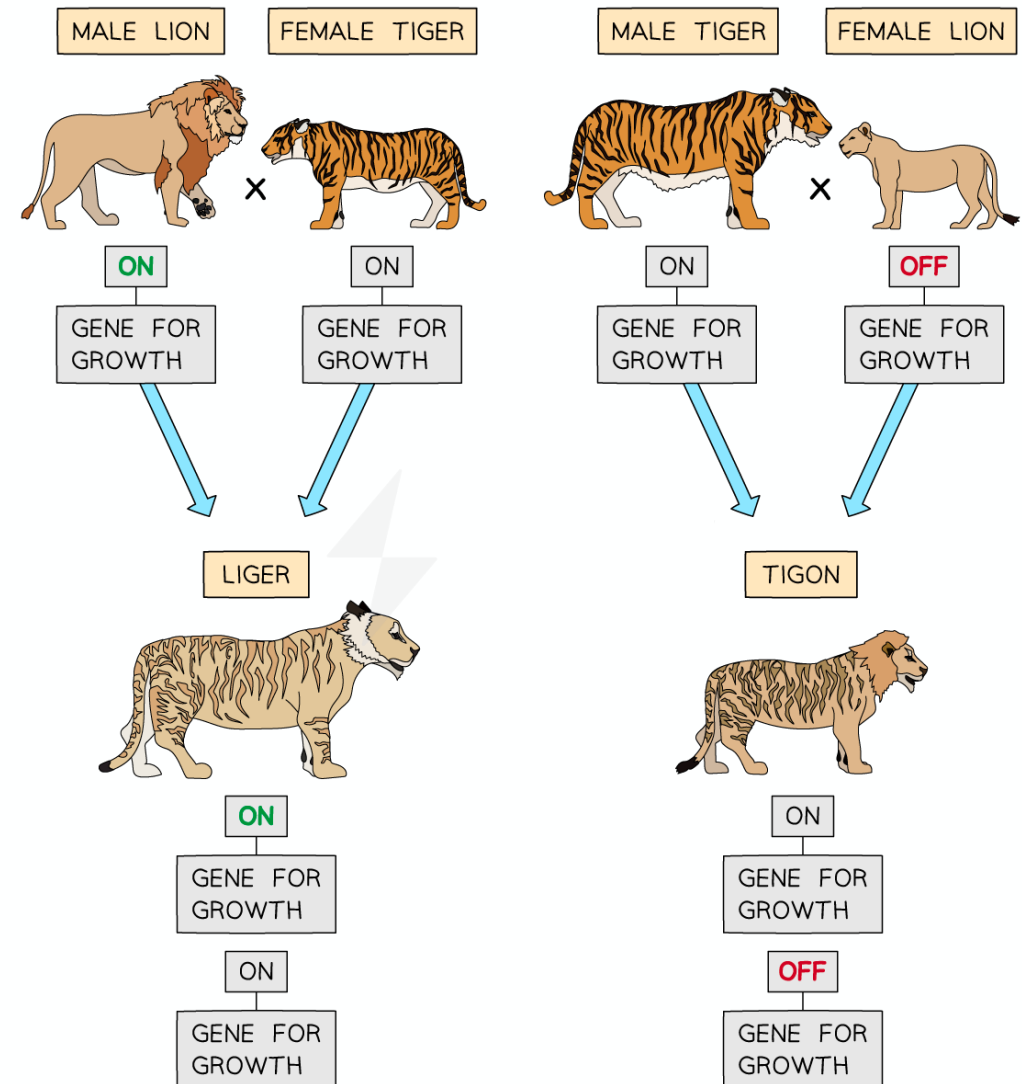


Genomic imprinting: mammalian parental tug-of-war

Imprinting has evolved in mammals because of the conflicting interests of maternal and paternal genes in relation to the *transfer of nutrients from the mother to her offspring*



Genomic imprinting: mammalian parental tug-of-war



Genomic imprinting: mammalian parental tug-of-war

How applicable to humans?

- In **mice**, knocking out the **paternal Igf2** allele causes severe growth restriction in utero
- Knocking out the **maternal allele** has no effect, confirming only the paternal allele is normally active.

DeChiara et al., 1991

paternal gene is imprinted?

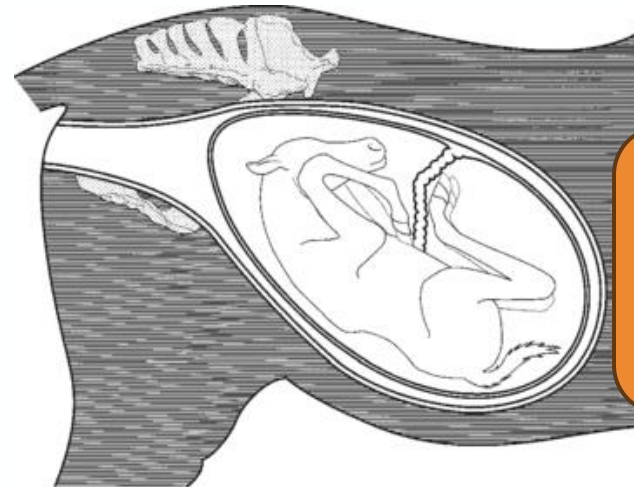


Paternally expressed Igf2 drives **fetal growth**



- In **equines**, many of the paternally expressed genes identified are known to support the development and function of the **placenta**—e.g., INSR (insulin receptor), which works alongside IGF2 and is involved in nutrient uptake and growth regulation

Wang et al., 2013



Many paternally expressed genes drive **placental growth**



Genomic imprinting: mammalian parental tug-of-war

In rodents, paternal gene influences on mother include:

- Increased feeding
- Decrease sexual desire
- Nest building

Postpartum:

- Maternal care
- Milk let-down

And in the newborn:

- Suckling



Behaviour can affect the epigenome

Figure 35.3

High-interaction mother

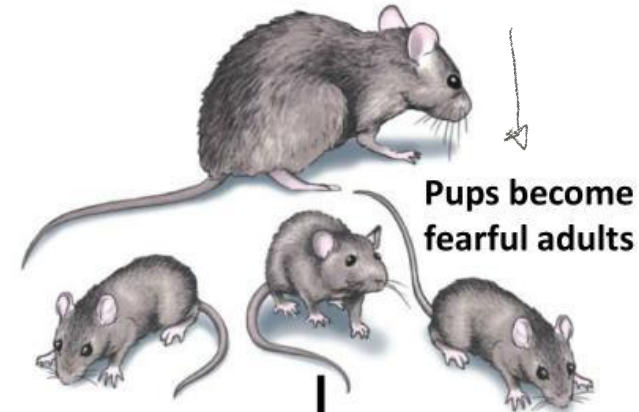


Female pups become high-interaction mothers



Pups become relaxed adults

Low-interaction mother

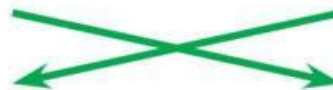


Female pups become low-interaction mothers



Pups become fearful adults

Cross-fostering experiment

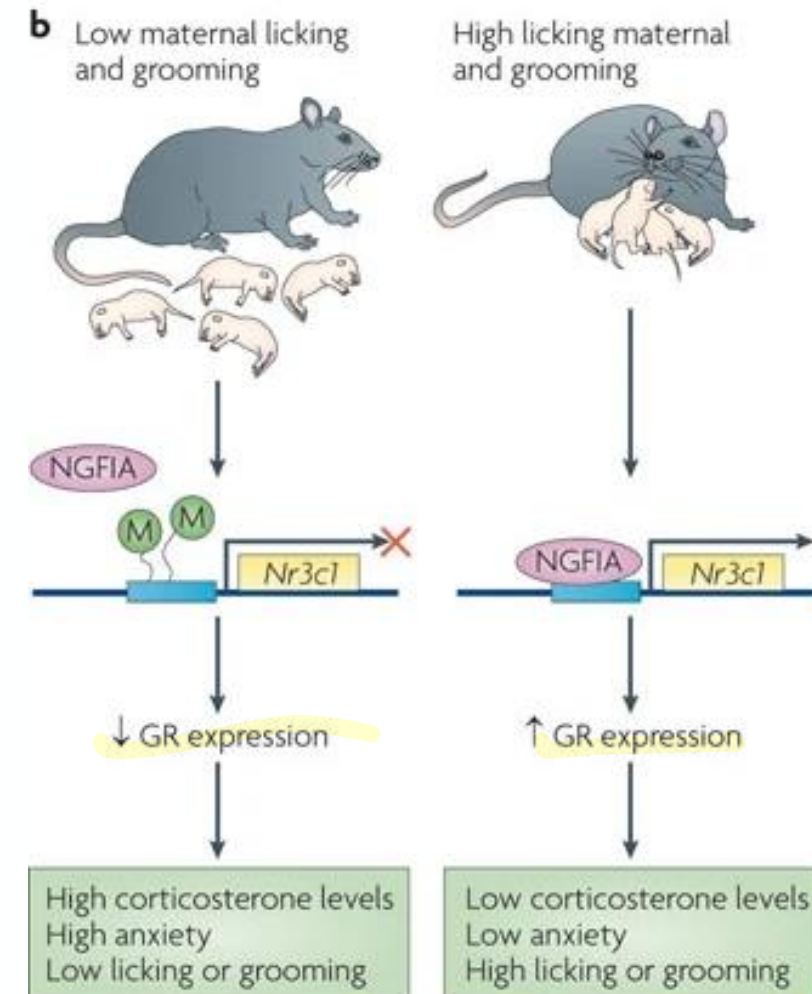


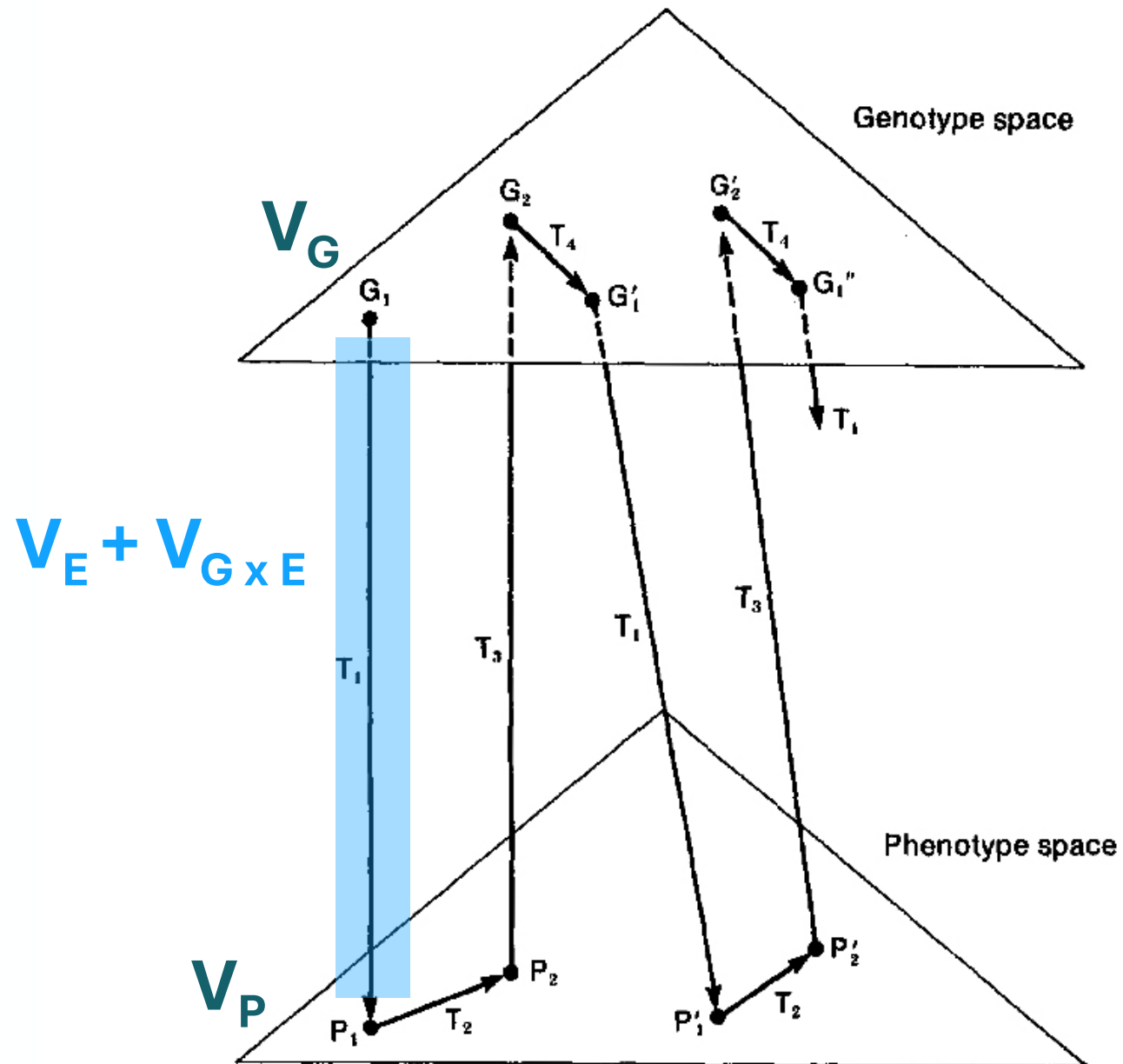
Behaviour can affect the epigenome (and thus influence phenotype in the absence of genetic change)

- Glucocorticoid receptors (GR) in the hippocampus mop up cortisol and turn off the stress response.
- Anxious rats have fewer GR



Maternal behavior influenced behavior of GR receptors - licking increases





1

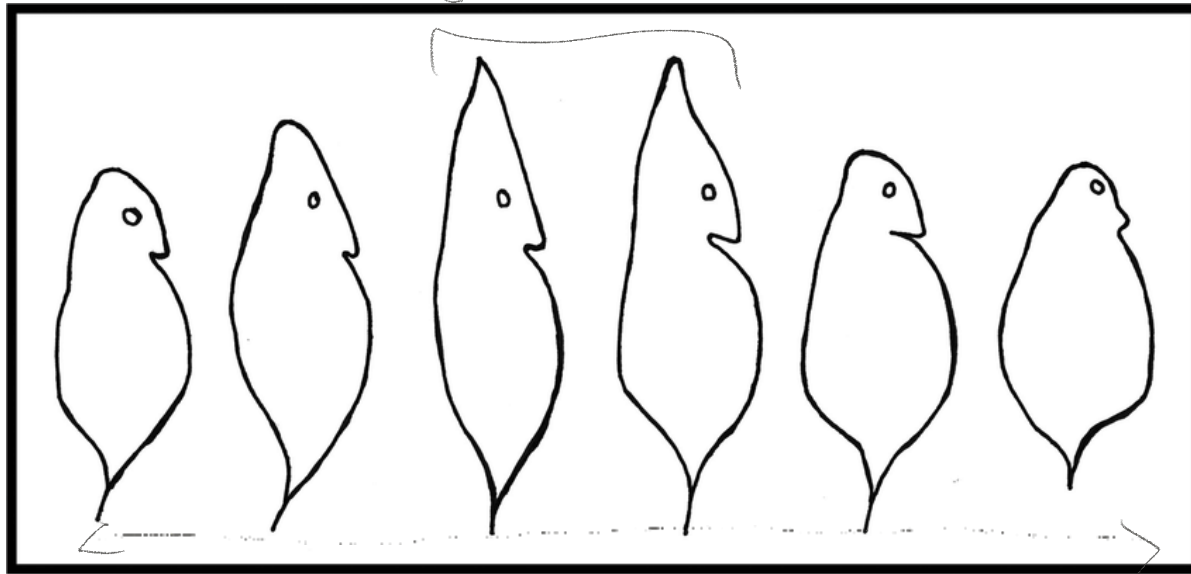
Epigenetics

2

Reaction
norms

Phenotypic plasticity generates phenotypic variation in the absence of genetic variation

summer - high predation → detected chemically?
[adaptive response]

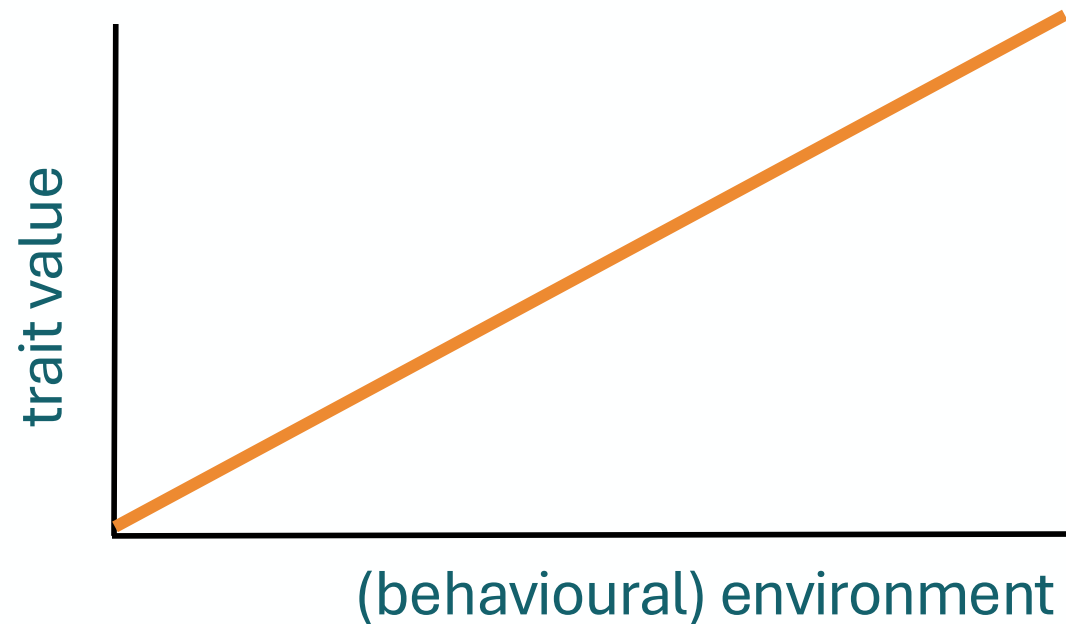


Woltereck, 1909: *Daphnia*



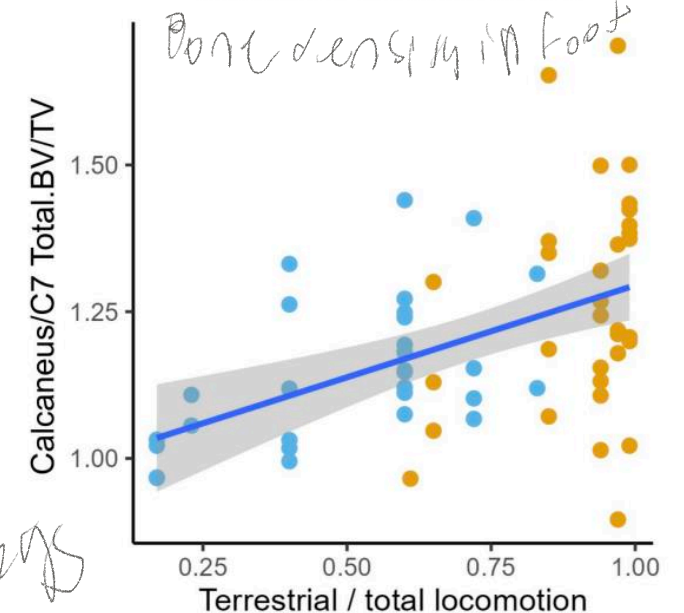
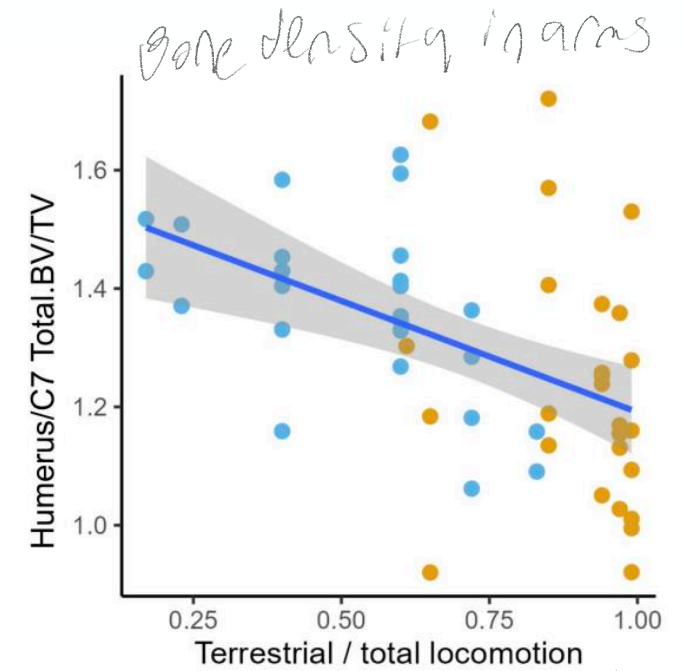
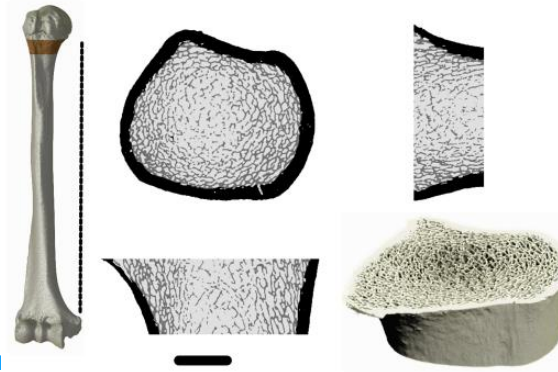
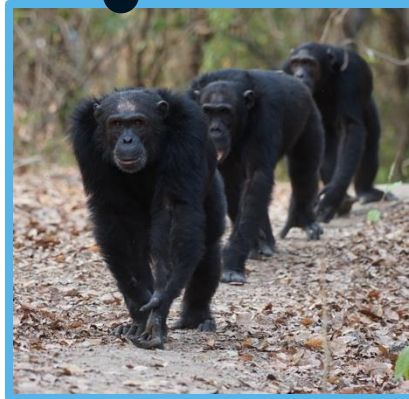
Phenotypic plasticity generates phenotypic variation in the absence of genetic variation

Reaction norms



Phenotypic plasticity in great ape bone density

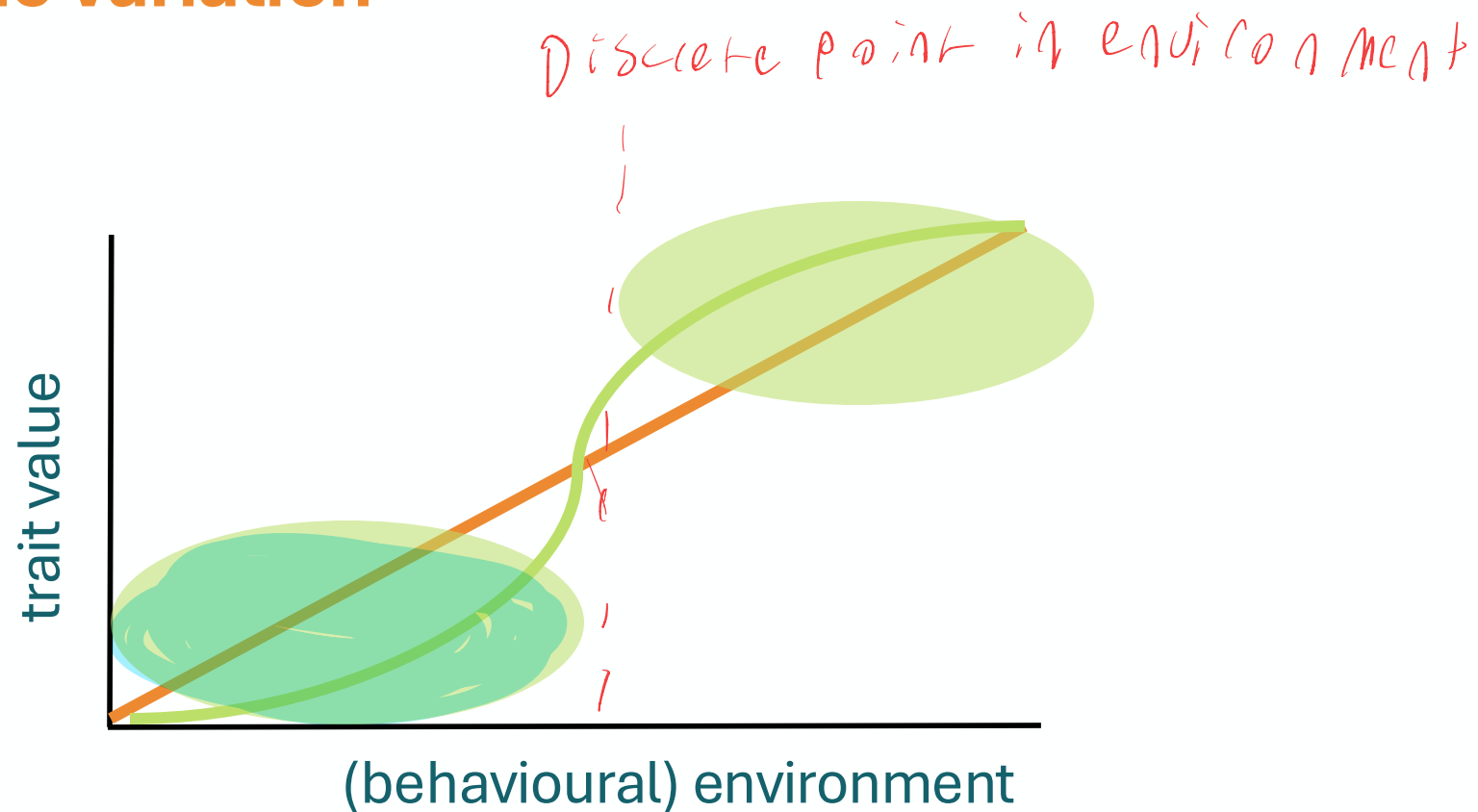
Example of a reaction norm



Babies are arboreal → No reloading on arms & legs
Terrestriality → loading on arms AND legs

Phenotypic plasticity generates phenotypic variation in the absence of genetic variation

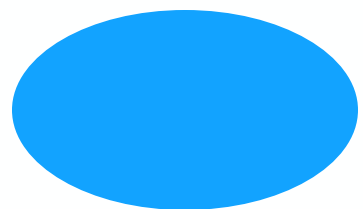
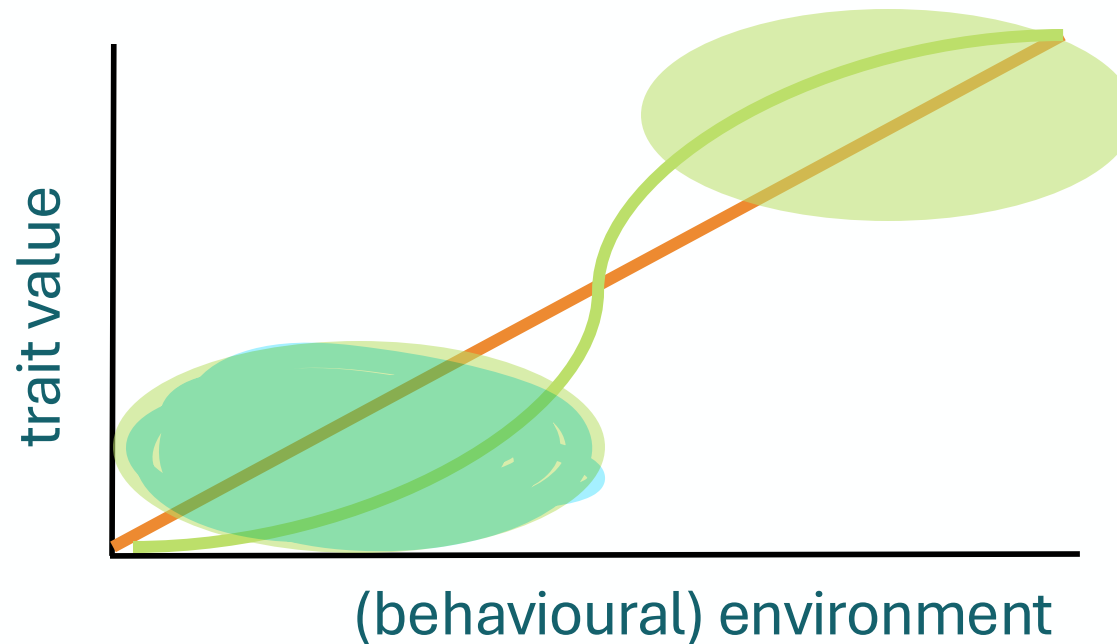
Reaction norms



polyphenism: discrete morphs

Phenotypic plasticity can influence selection and local genetic diversity

Reaction norms

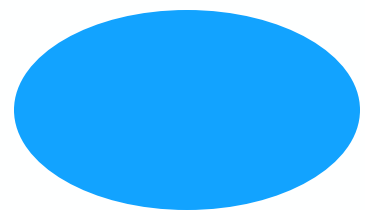
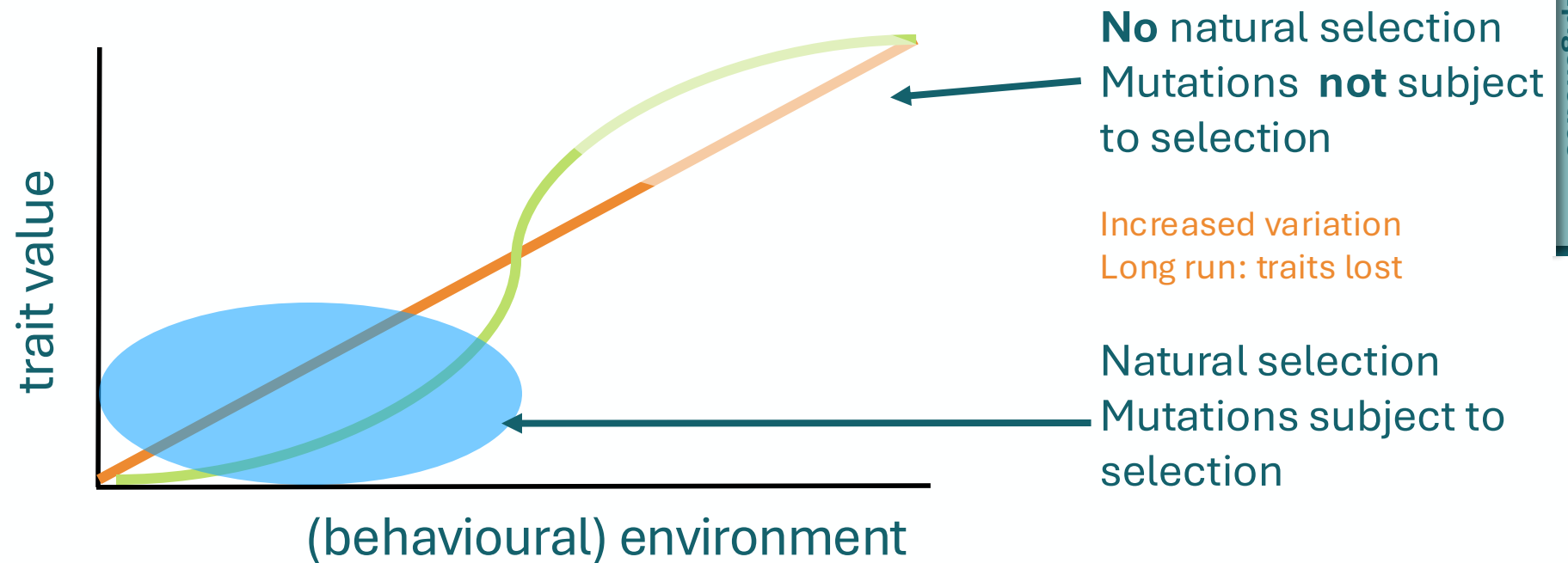


Genes associated with blue phenotype are **expressed**

Genes associated with white phenotype are **not expressed**

Phenotypic plasticity can influence selection and local genetic diversity

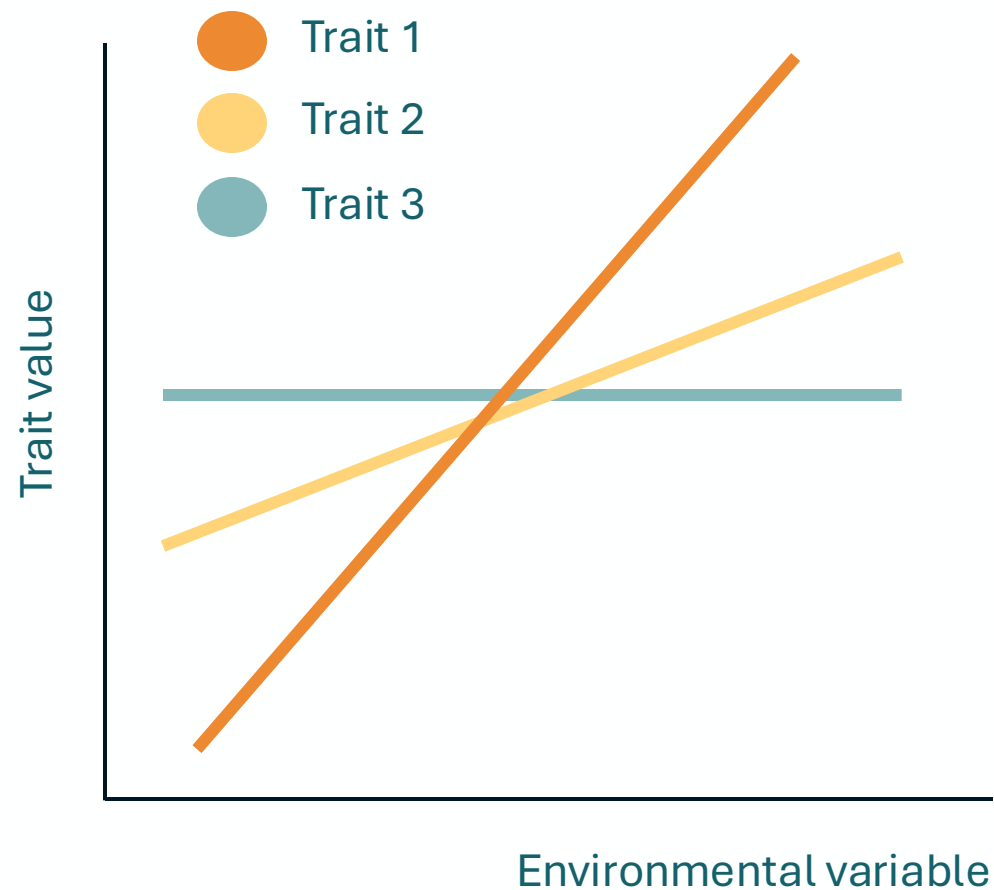
Reaction norms



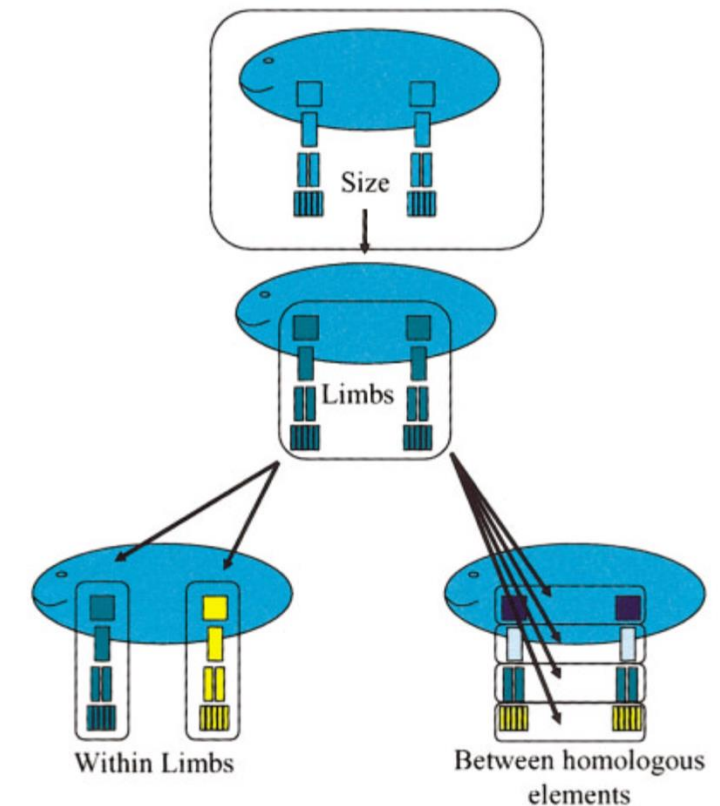
Genes associated with blue phenotype are **expressed**

Genes associated with white phenotype are **not expressed**

It is useful to think of the organism as a mosaic of sensitivities.

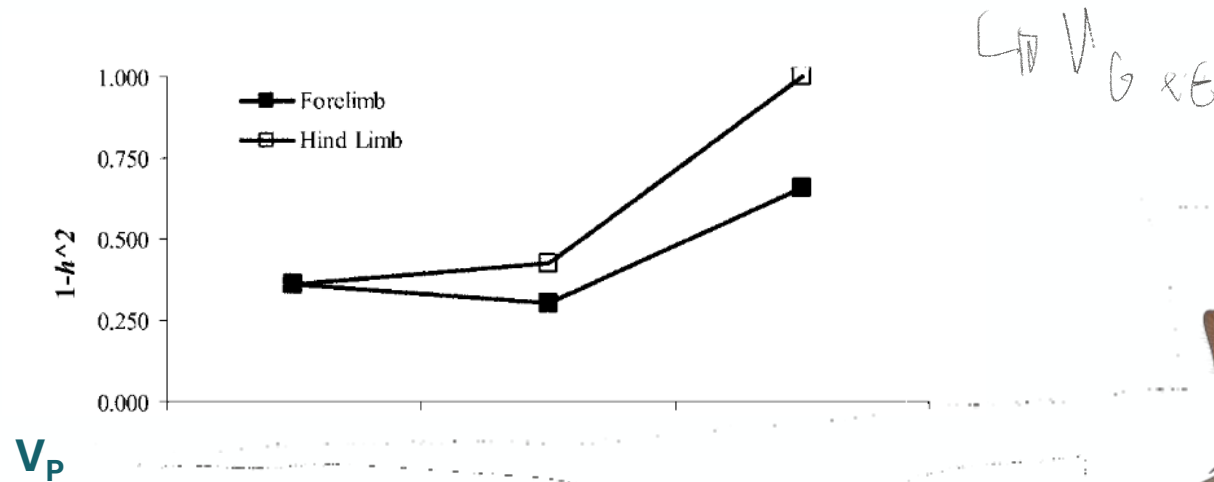


Different traits react differently depending on environment

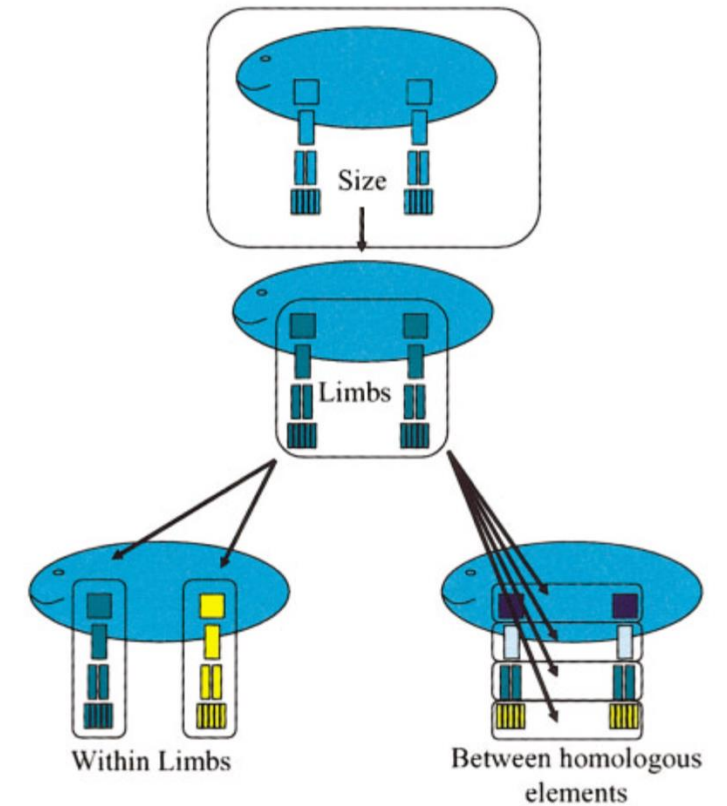
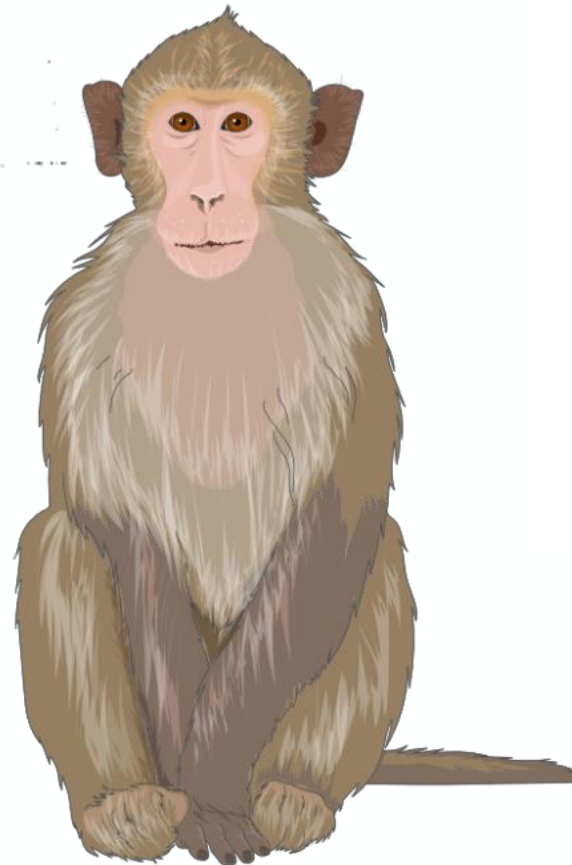
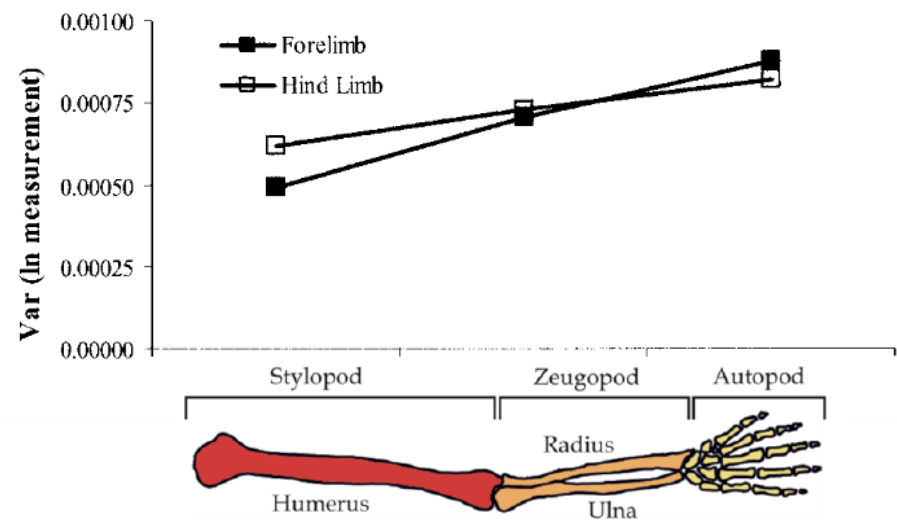


It is useful to think of the organism as a mosaic of sensitivities.

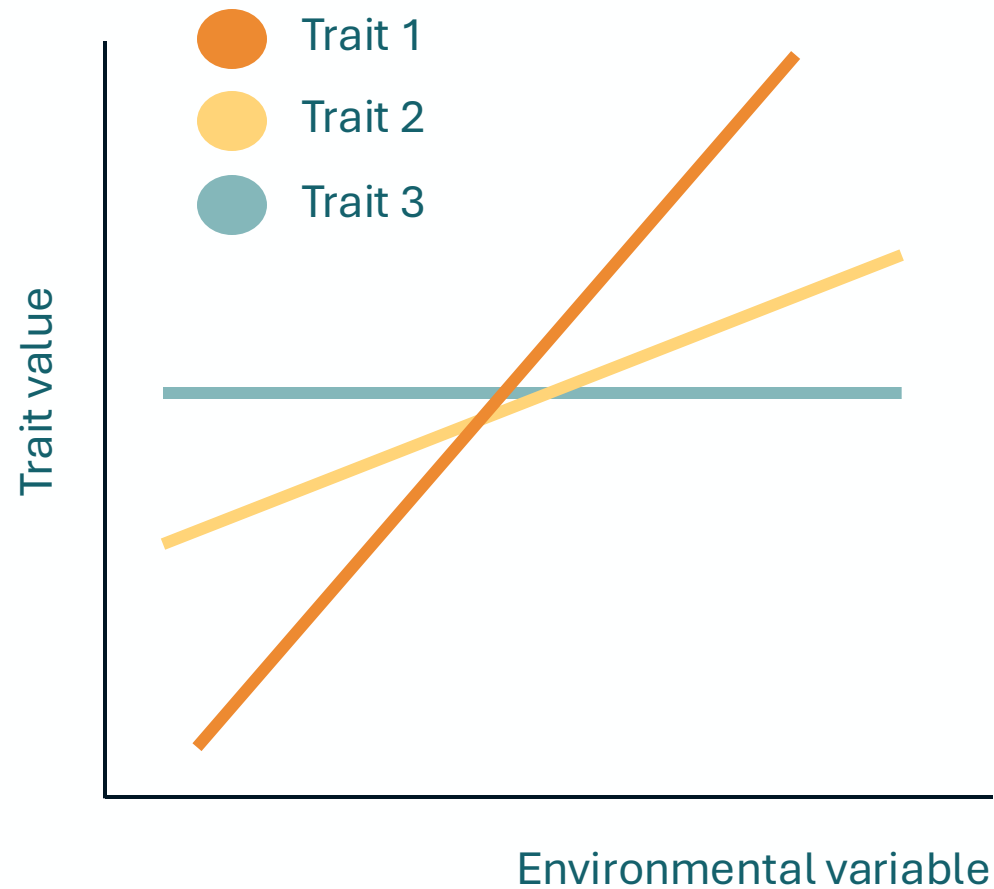
The portion of V_P due to variation in the environment



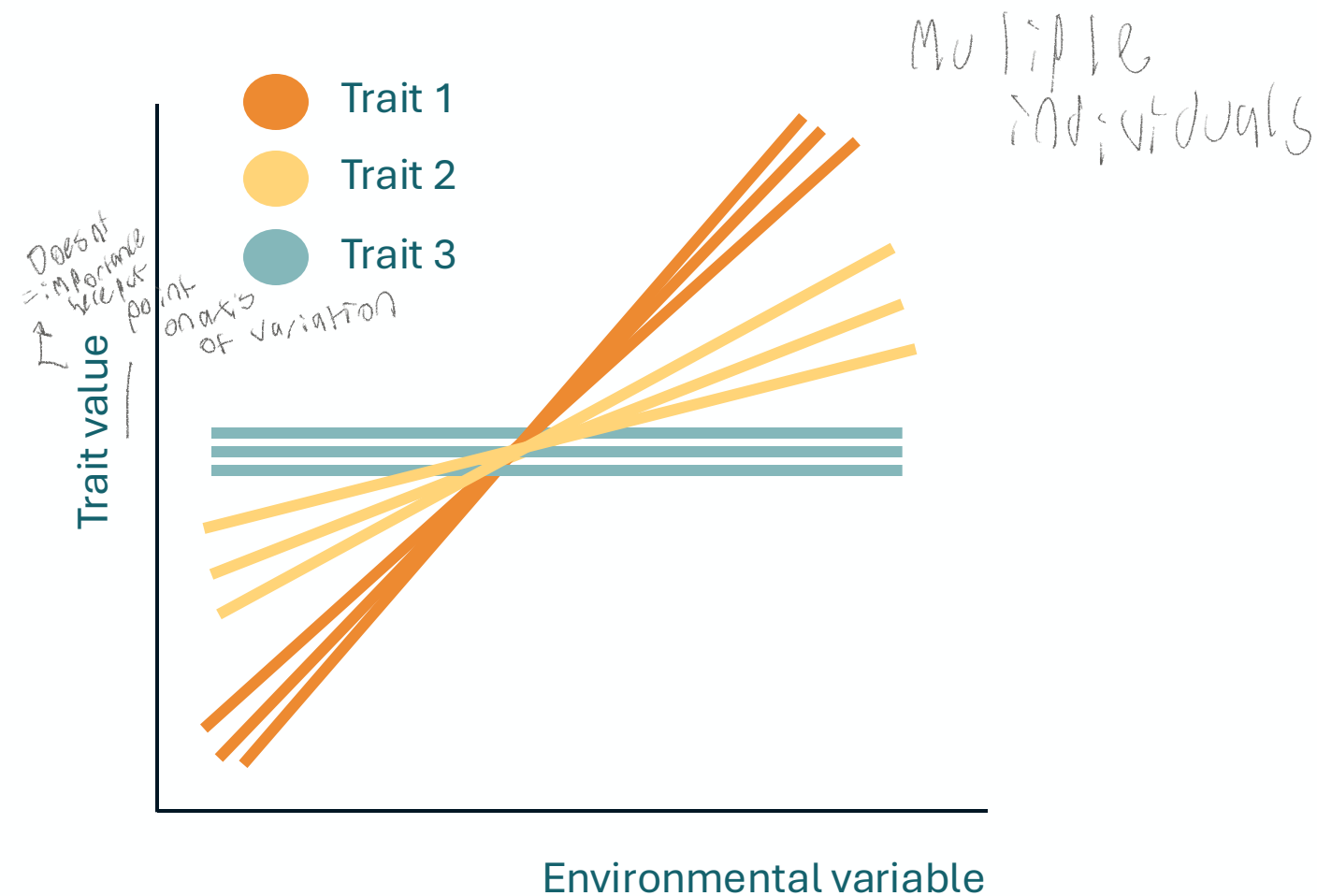
V_P



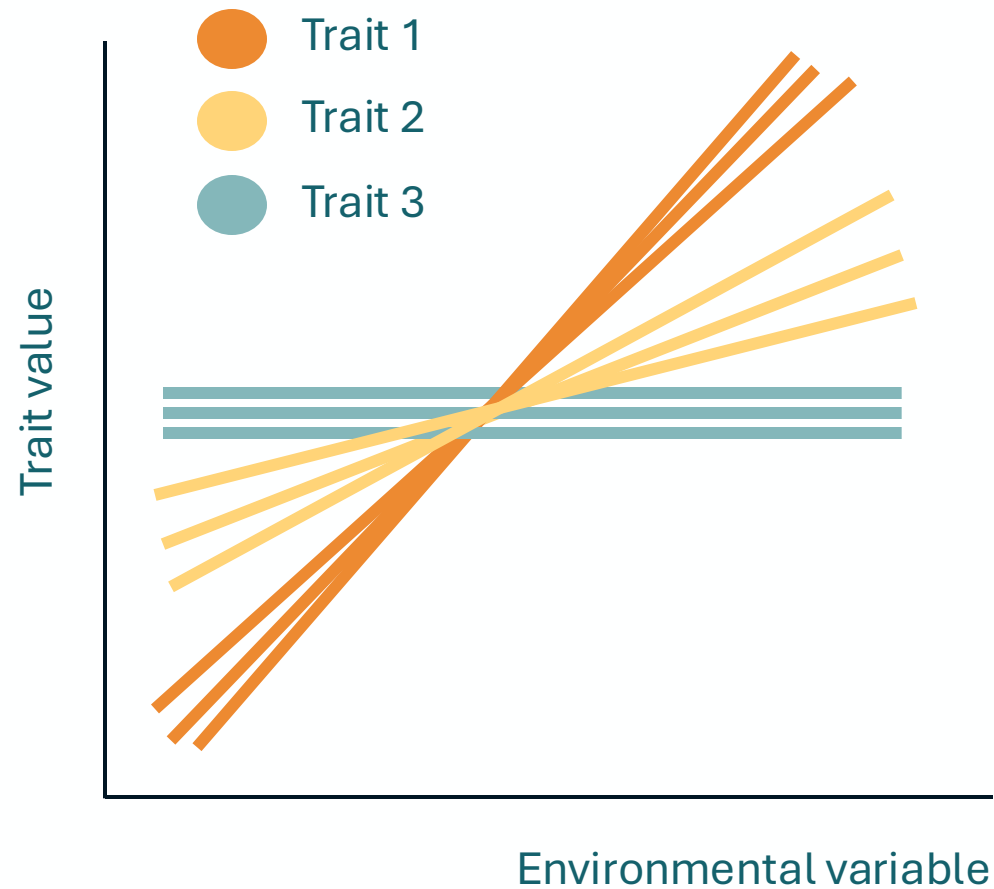
It is useful to think of the organism as a mosaic of sensitivities.



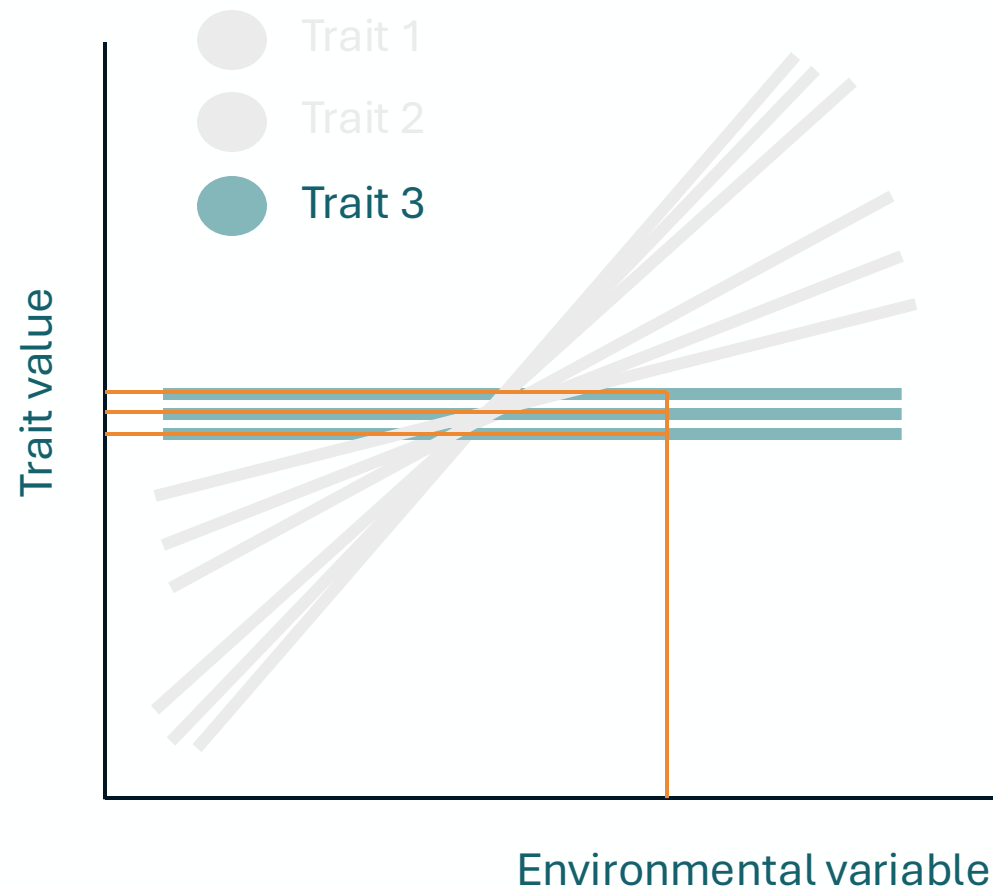
It is *also* useful to think of a population as a group of sensitivities.



‘Pub’ quiz! Get yourselves into groups and think of a group name!



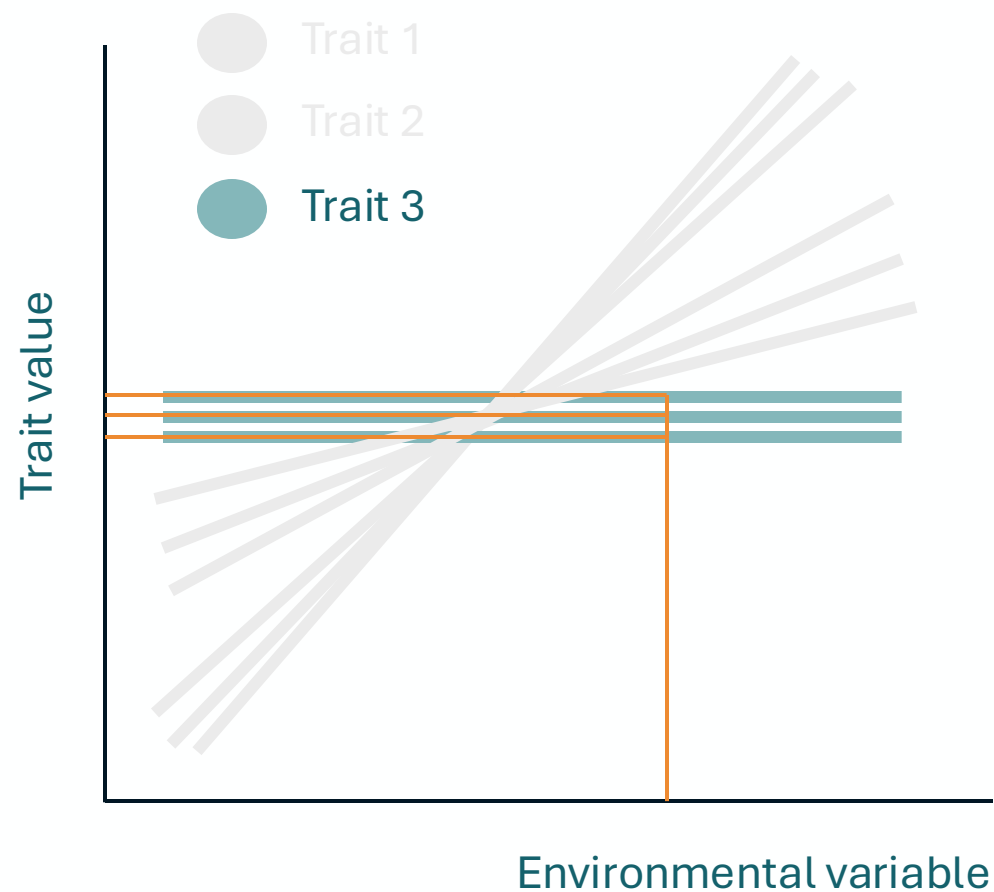
1. Which of the following statements is correct?



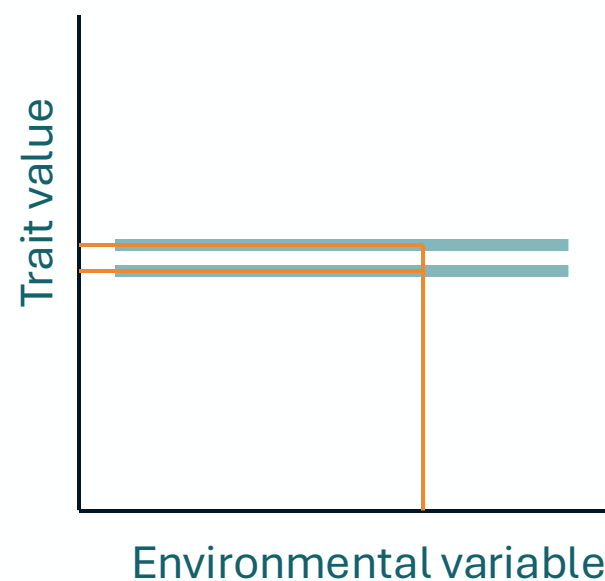
$$V_P = V_G + V_E + V_{G \times E}$$

1. $V_G = 0$. All of the differences between people in this trait are due to the environment.
2. $V_G = 1$. All of the differences between people in this trait are due to genetic differences between them.
3. $V_{G \times E} = 1$. All of the differences between people are due to genotype x environment interactions.

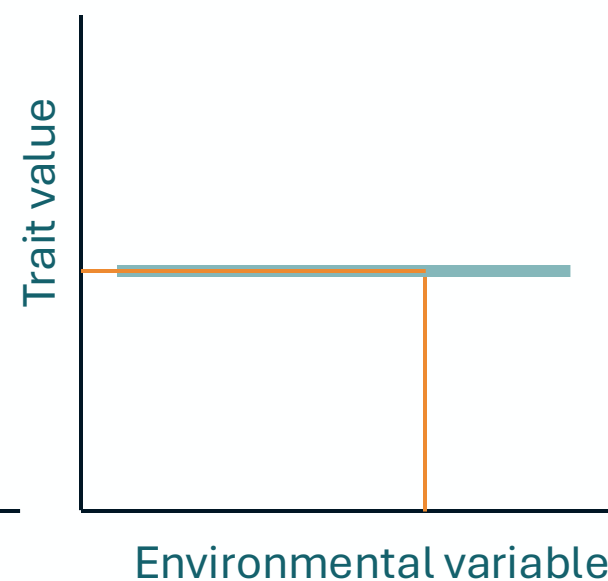
2. Which of the following drawings is correct for this trait for monozygotic twins?



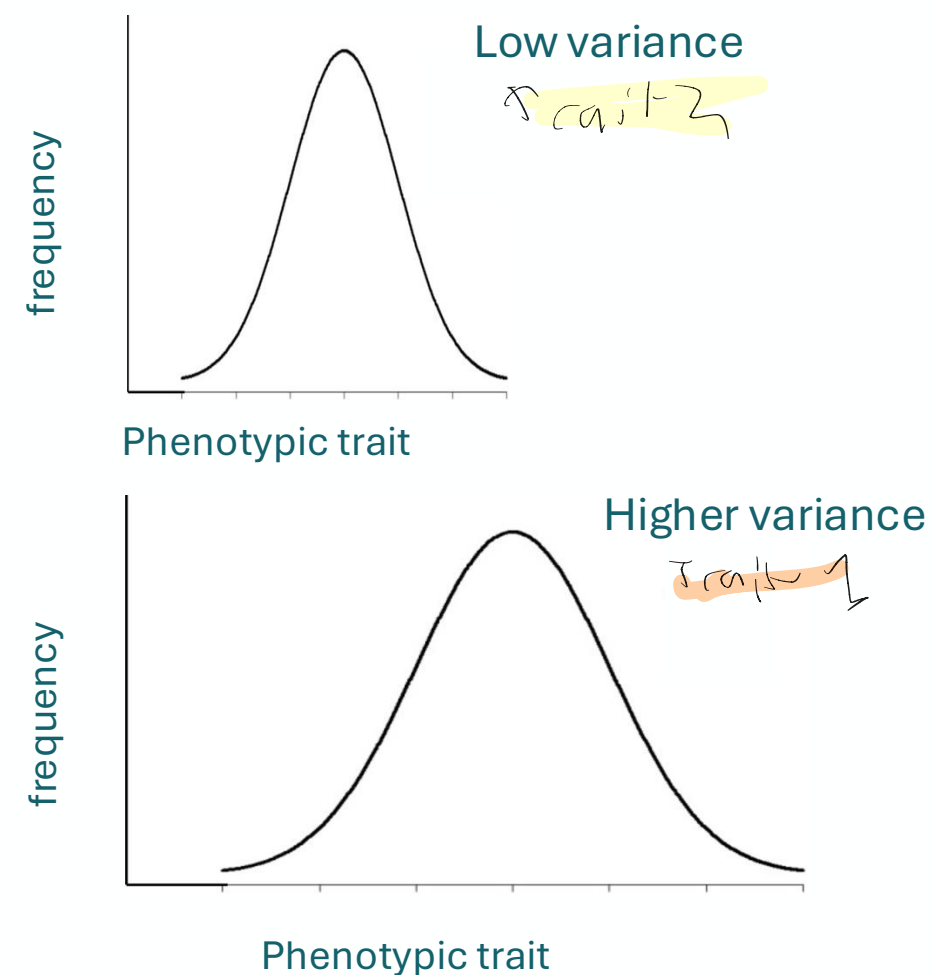
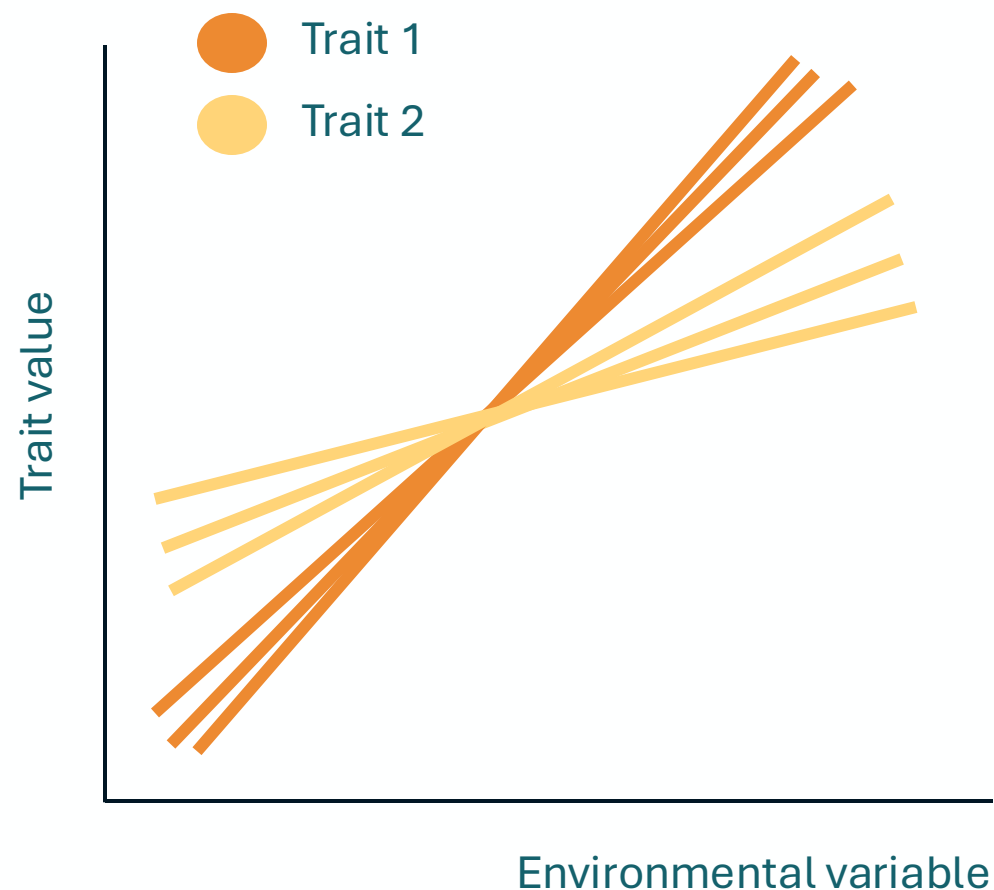
a)



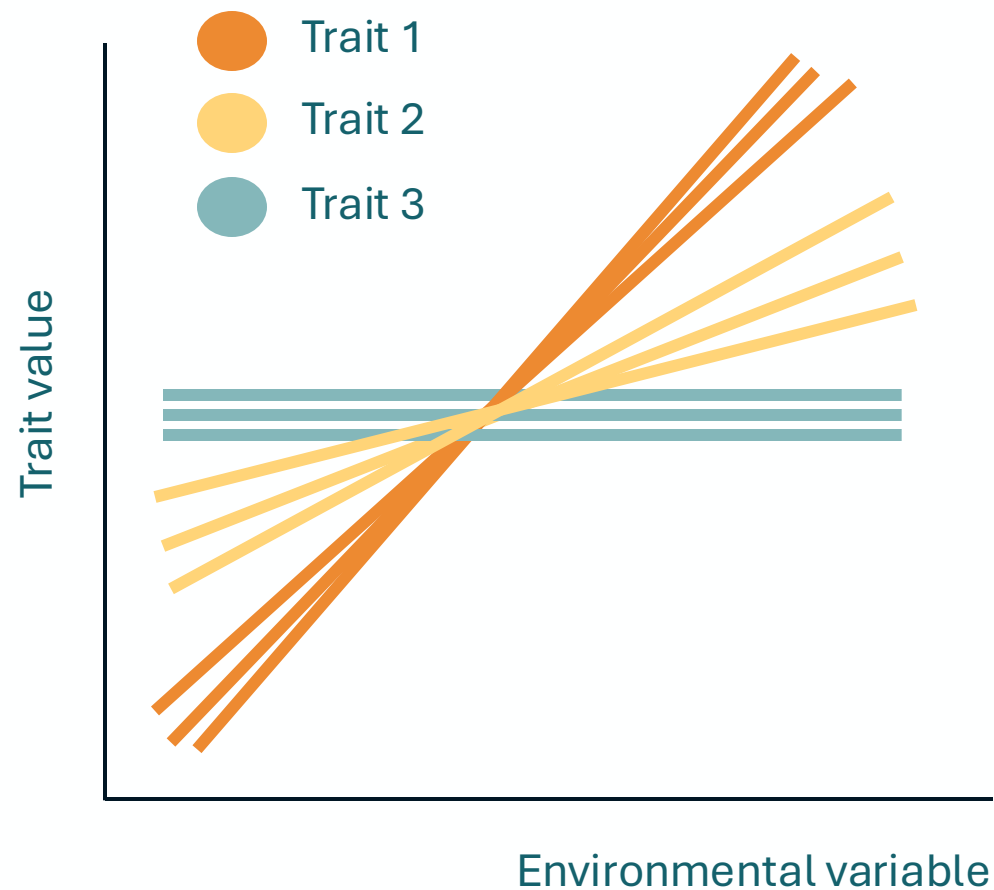
b) only 2 genotype



3. Do you expect that, in a population of people exposed to different environments during ontogeny, Trait 1 or Trait 2 would have higher variance?



4. Order traits 1-3 by likelihood of being polygenic.



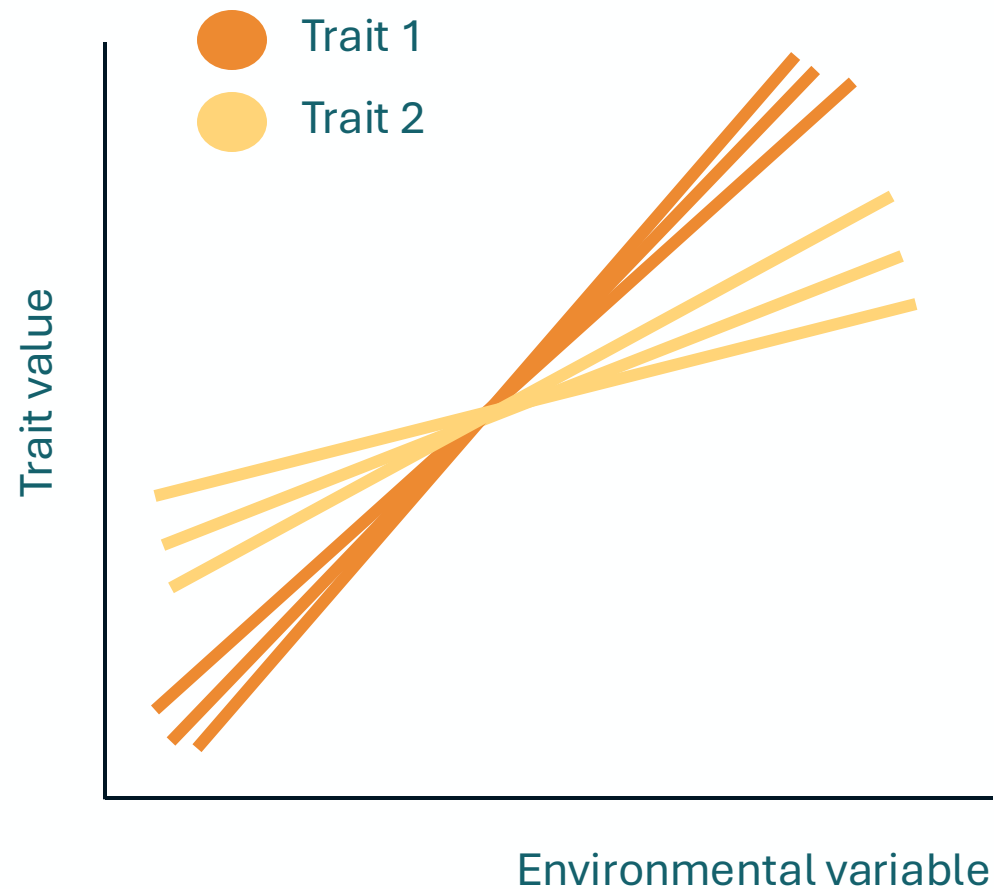
most likely to be polygenic
Trait 2
Trait 2
Trait 3
most likely to be monogenic



Single gene: Large effect each

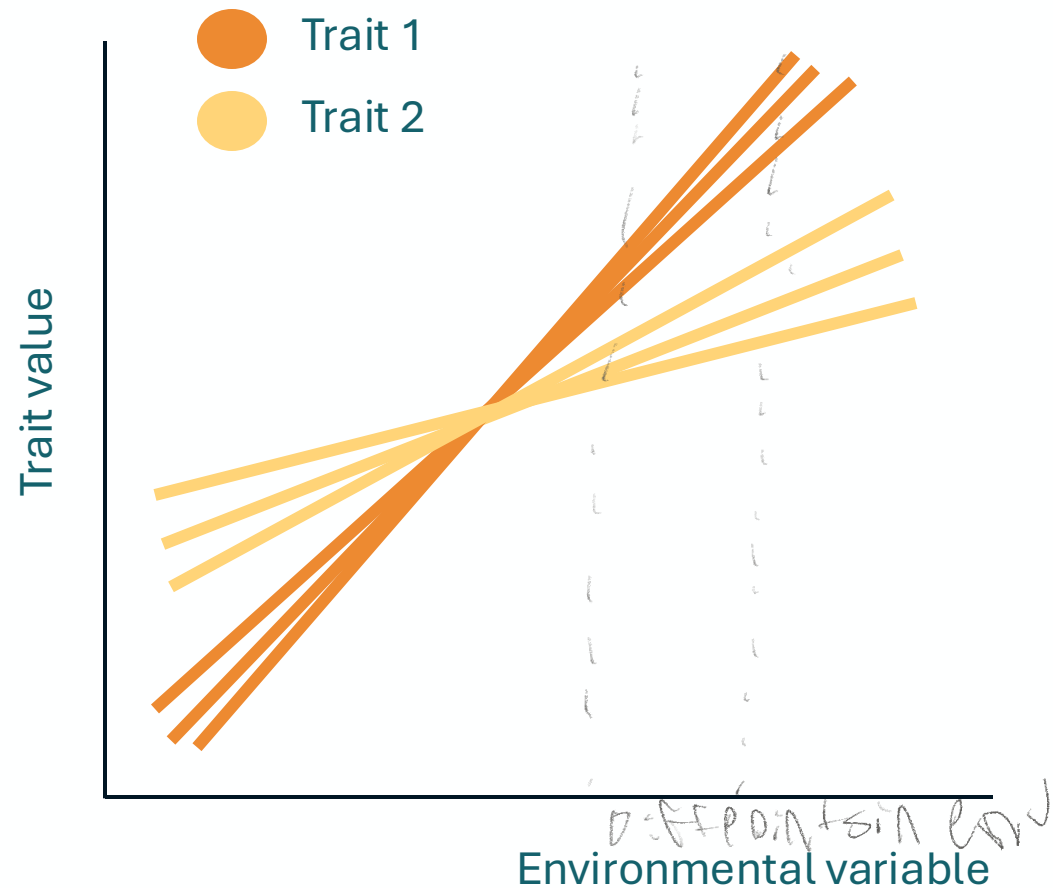
Polygenic: Small effect each

5. Which of the following statements is correct?



1. More of the variance in trait 2 is due to gene x environment interactions. We therefore expect twins living in different environments to differ more to each other in trait 2 than trait 1.
2. More of the variance in trait 1 is due to gene x environment interactions. We therefore expect twins living in different environments to differ more to each other in trait 2 than trait 1.
3. More of the variance in trait 1 is due to gene x environment interactions. We therefore expect twins living in different environments to differ more to each other in trait 1 than trait 2.
4. More of the variance in trait 2 is due to gene x environment interactions. We therefore expect twins living in different environments to differ more to each other in trait 1 than trait 2.

6. Which human traits would you expect to follow the pattern in trait 1? What about trait 2?



- a) Adult height, skin colour, fat distribution $\leftarrow$ Trait 1
- b) Body mass, muscle vs fat ratio, pesticide-induced Parkinson's disease $\leftarrow$ Trait 2

KEY

Genes set the limit
of phenotype, environment
determines whether limits
are reached

Important reminders

1. Date & time of final exam has been confirmed: **12/11/2025; 8-10.30AM**
2. Midterm exam: **10/16/2025 during class**
3. Revision session for midterm exam: **next week Thursday**
4. Assignment coming up! Due: **10/10/2025 11.59PM**

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 - **Choice between Adaptation Matrix OR Red List**
 - **All information is in syllabus and on Canvas**

The screenshot shows the Canvas LMS interface for the course FA25_ANT_210_1. The assignment is titled 'Adaptation Matrix OR Red List' and is currently published. The instructions are as follows:

INSTRUCTIONS:

1. Choose which assignment you'd like to complete: the Adaptation Matrix or the Red List.
2. Read the assignment instructions for your chosen assignment, which can be found in the most up-to-date syllabus, carefully.
3. Download fillable Adaptation Matrix [here](#), and the fillable Red List form [here](#).
4. Fill in the appropriate document for your chosen assignment according to the instructions.
5. Upload your work as PDF, with the file name specified on the Adaptation Matrix or Red List document, here on Canvas **before the due date and time (Friday 10 October at 11.59PM)**.
6. Late submissions are **not permitted**, unless previous arrangements have been made directly with Dr. Van Holstein *at least 48 hours before the deadline*.

Submission details: Points: 5, Submitting: a file upload.

Due	For	Available from	Until
Oct 10	Everyone	-	Oct 11 at 12:30am

Additional options: + Rubric

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EMORY
FA25_ANT_210_1 > Assignments > Adaptation Matrix OR Red List

Fall 2025

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Grades
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Gradescope
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Collaborations
Rubrics

Adaptation Matrix OR Red List

Published Assign To Edit

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Points 5
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